

Introduction:

Basketball is a sport that relies on many numerical records to quantify a player's or team's performance. One question that may be posed is whether these statistics, as a whole, accurately predict the outcome of a game. Furthermore, we may question whether some individual statistics represent the outcome of a game better than others. These questions can be answered through a multiple linear regression model. First, we can use an ANOVA F-Test to assess the cumulative effect of all predictors. Then we can utilize t-tests, confidence intervals on the slope, and best-subsets to examine the individual contribution of each variable. The response in this analysis is point differential, as it provides a continuous variable that directly determines whether a team performed excellently or poorly relative to its opponent. The explanatory variables are, of course, the numerical records of the games. Based on my individual experience playing and watching basketball, I would predict that field goals made, offensive rebounds, and defensive rebounds would have the strongest positive correlation with point differential, and that turnovers, personal fouls, and blocks against would have the strongest negative correlation. Overall, this model should offer useful insights regarding the importance and effectiveness of basketball statistics. These insights could help a team determine which aspects of the game most influence their chance of winning — and are therefore most important to their success.

Analysis of the Explanatory Variables - Descriptive Statistics:

Statistics

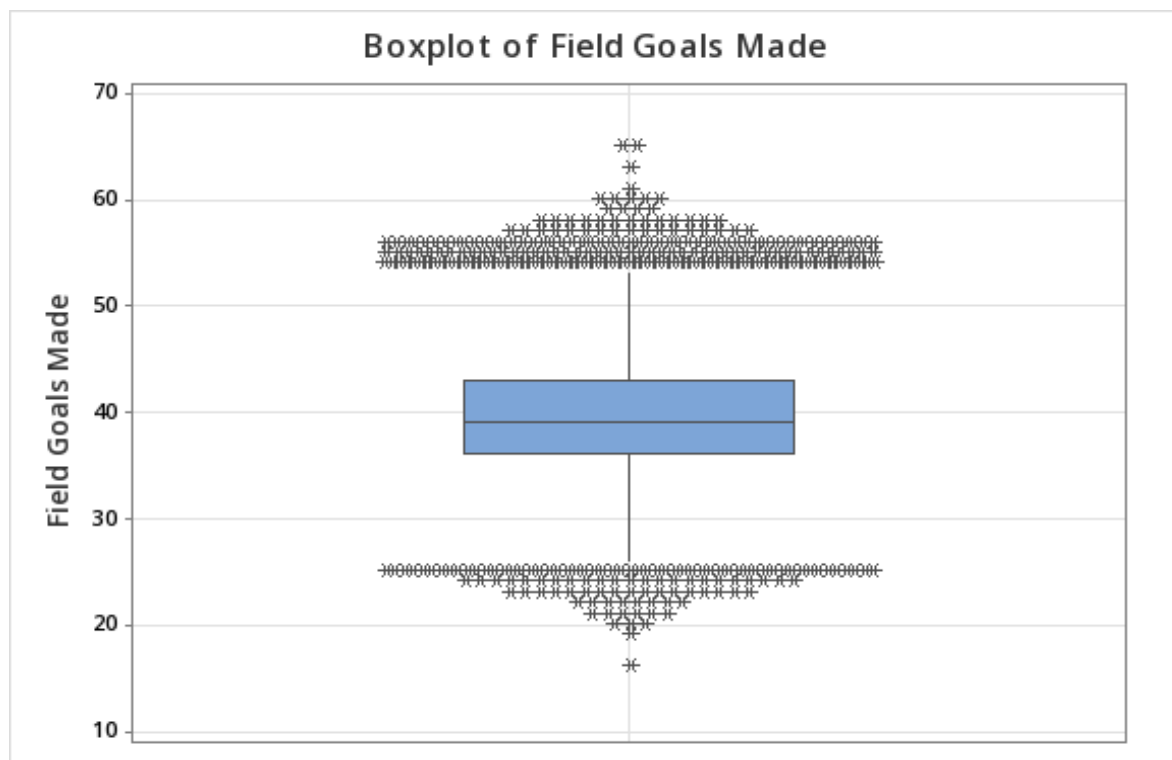
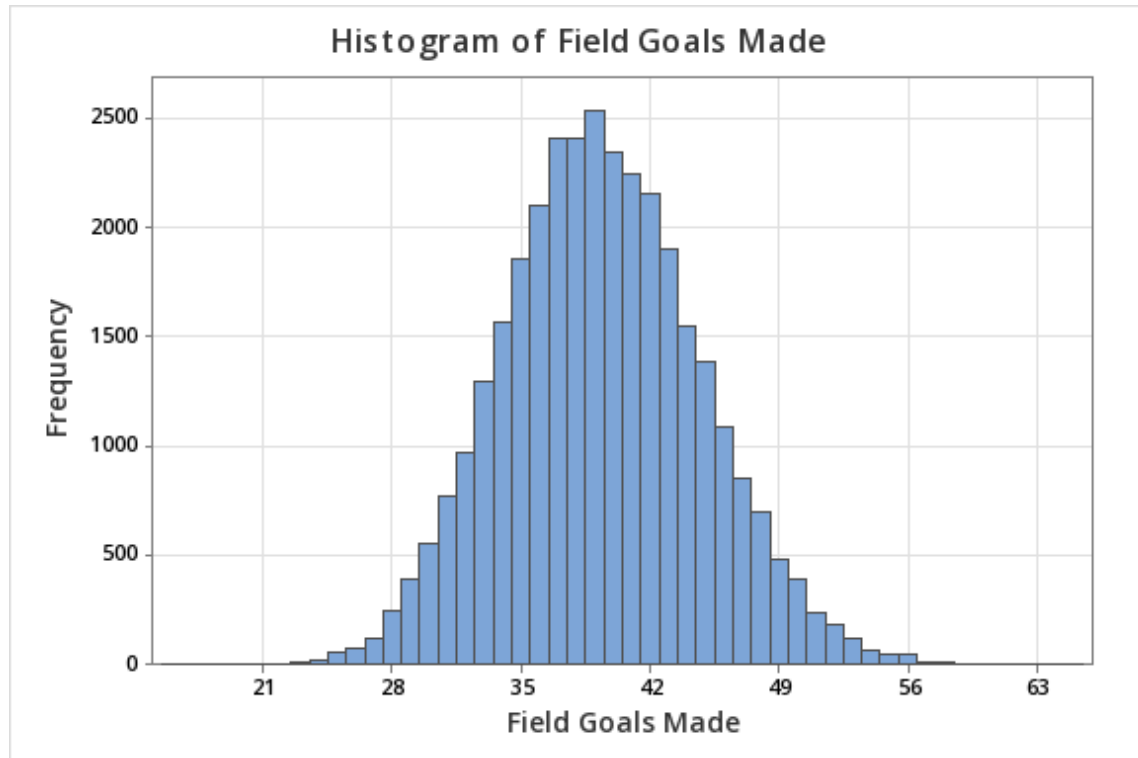
Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Field Goals Made	33316	0	39.3598	0.0295935	5.40160	16	36	39	43	65
Field Goals Attempted	33316	0	85.6476	0.0419640	7.65956	59	80	85	91	129
Field Goal %	33316	0	0.460423	0.0003045	0.0555794	0.246	0.422	0.459	0.5	0.687

3-Pointers Made	33316	0	9.86985	0.0233111	4.25489	0	7	10	13	29
3-Pointers Attempted	33316	0	27.5470	0.0501483	9.15340	3	21	27	34	70
Three-Point %	33316	0	0.355998	0.0005344	0.0975340	0	0.292	0.355	0.419	0.842
Free Throws Made	33316	0	17.4500	0.0328236	5.99119	0	13	17	21	52
Free Throws Attempted	33316	0	22.7742	0.0400692	7.31370	0	18	22	27	64
Free Throw %	33316	0	0.766747	0.0005659	0.103290	0	0.7	0.773	0.839	1
Offensive Rebounds	33316	0	10.5045	0.0210559	3.84327	0	8	10	13	38
Defensive Rebounds	33316	0	32.9393	0.0300483	5.48463	12	29	33	37	60
Total Rebounds	33316	0	43.4439	0.0361559	6.59941	17	39	43	48	81
Assists	33316	0	23.3839	0.0288551	5.26683	4	20	23	27	50
Turnovers	33316	0	14.2020	0.0213928	3.90476	1	11	14	17	31
Steals	33316	0	7.62366	0.0158845	2.89935	0	6	7	9	22
Blocks	33316	0	4.87904	0.0138905	2.53540	0	3	5	6	20
Blocks Against	33316	0	4.87904	0.0138905	2.53540	0	3	5	6	20
Personal Fouls	33316	0	20.0321	0.0235354	4.29583	4	17	20	23	42
Personal Fouls Drawn	33316	0	20.0320	0.0235355	4.29586	4	17	20	23	42
Points	33316	0	106.040	0.0744368	13.5867	56	97	106	115	176

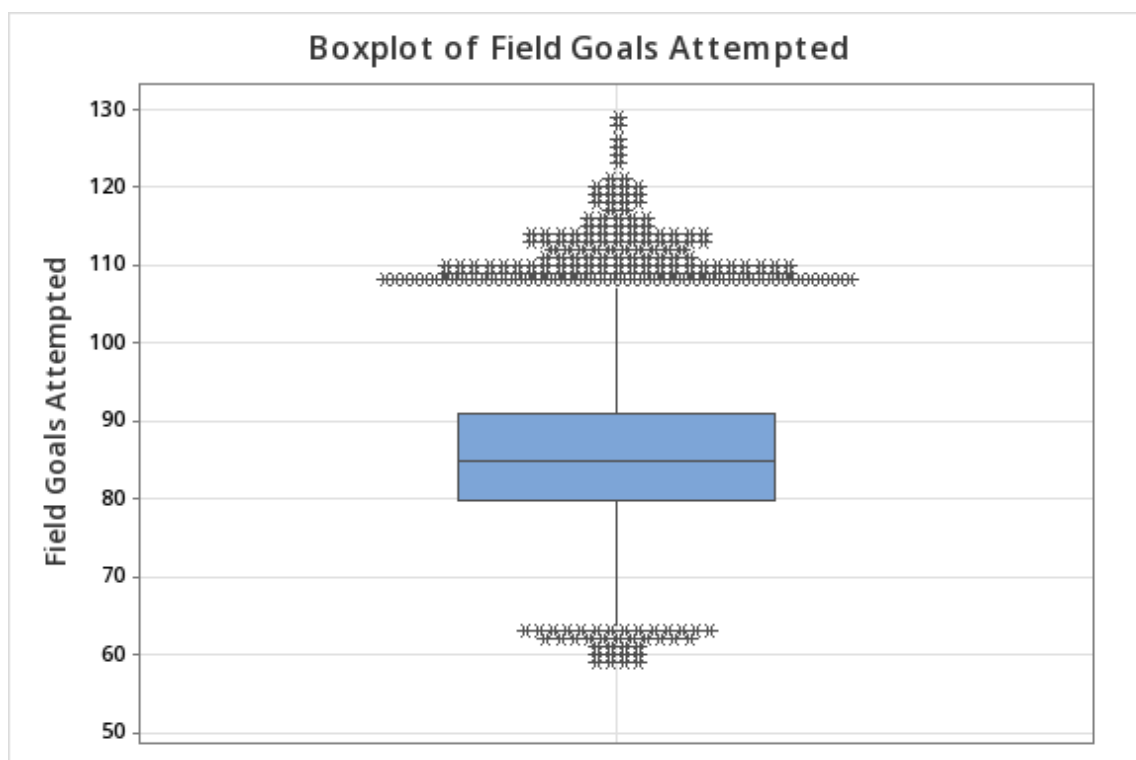
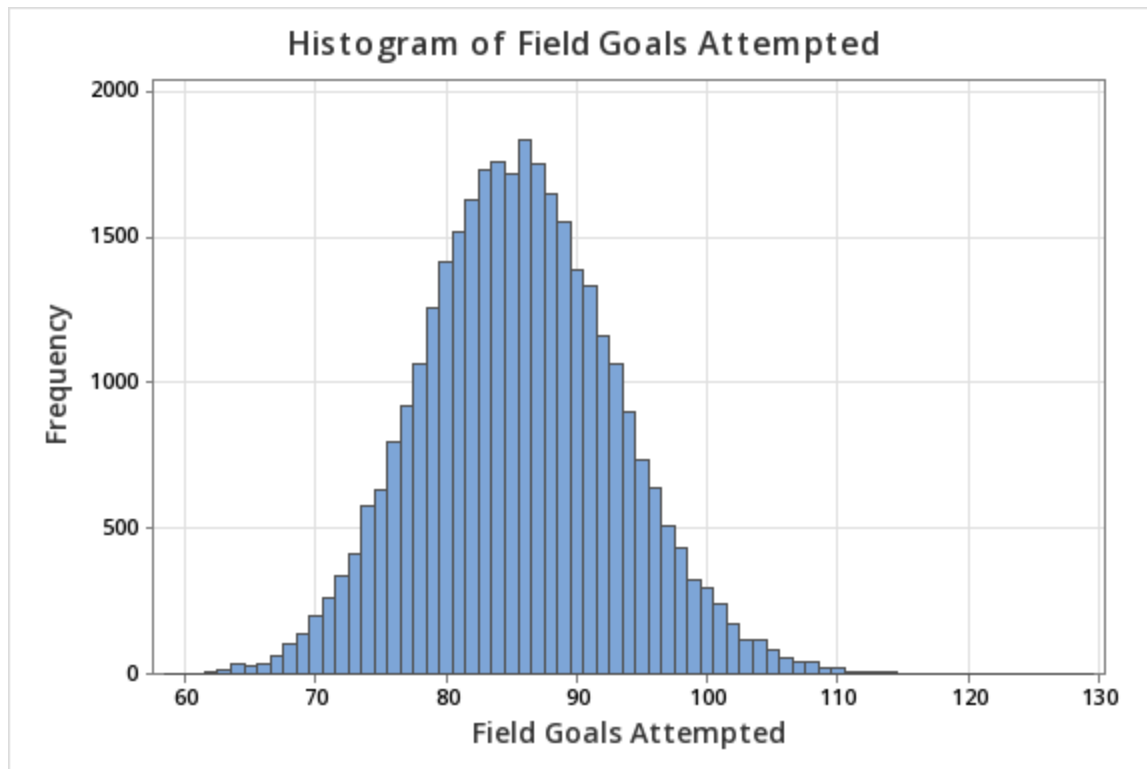
Summary of Basic Statistics:

In this output, N represents the sample size, which is the number of games recorded in the data set. N is the same for all explanatory variables as they are recorded for every game. The mean is the sum of the values divided by the number of values, yielding an expected value if randomly selected in the long run. The standard deviation and the standard error are different ways of representing the variability of the data. Finally, the five number summary chunks up the data into the bottom 25% (lower than Q1), top 25% (higher than Q3), and middle 50% (Q1 to Q3).

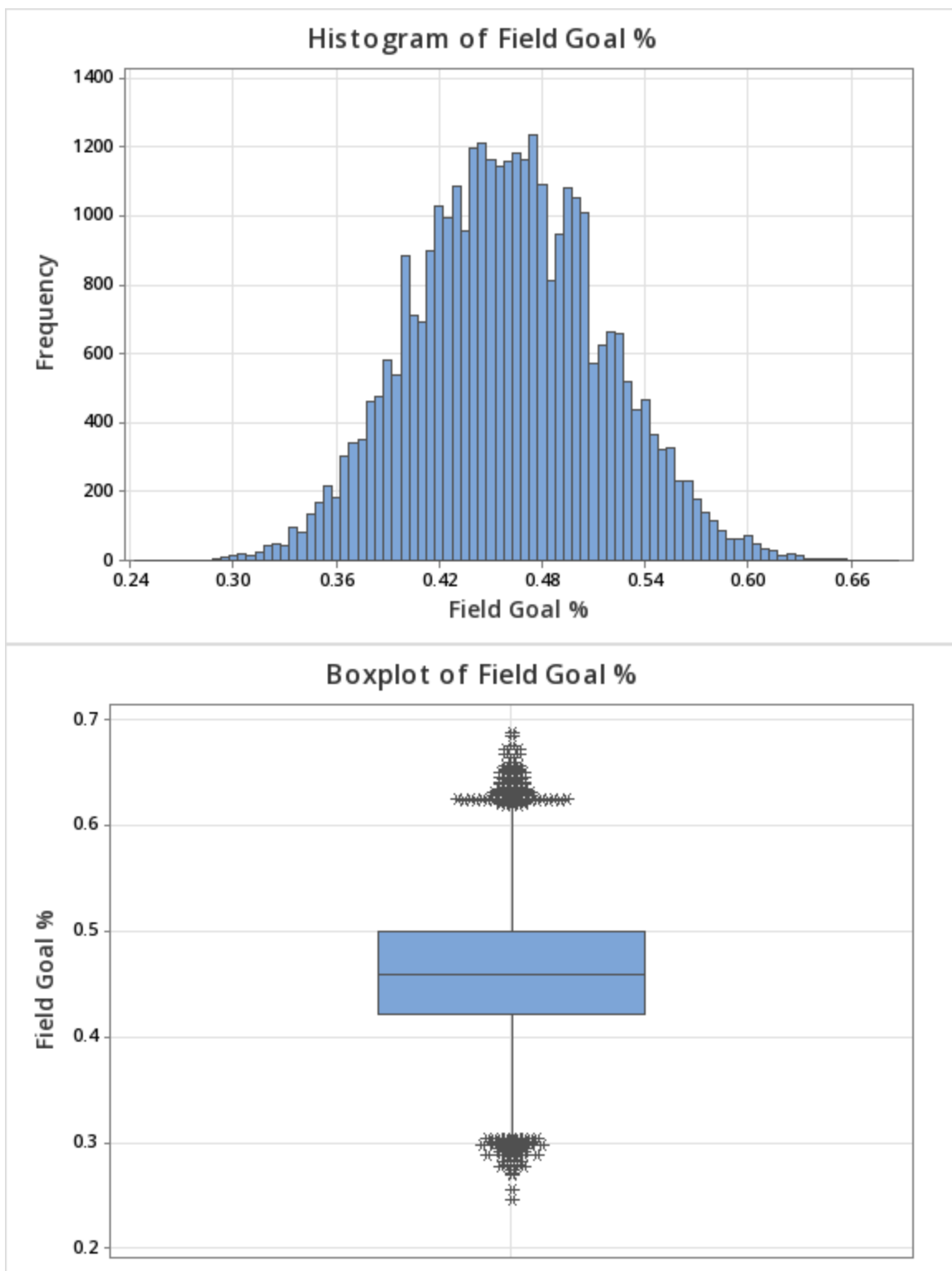
Explanatory Variable Distributions:



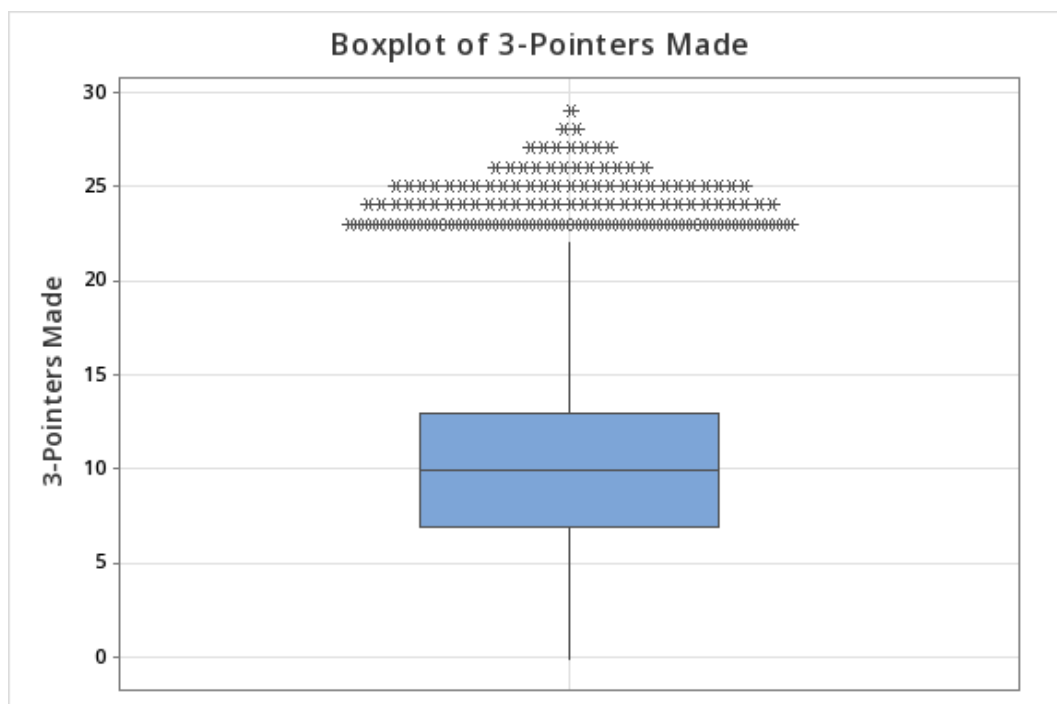
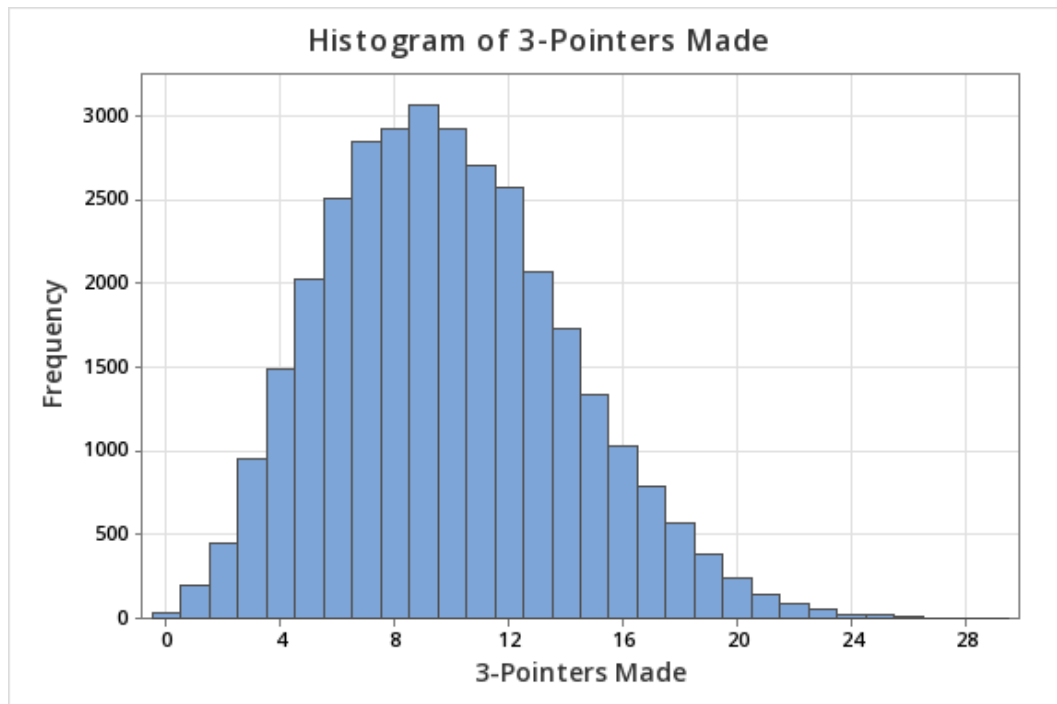
The boxplot and histogram of field goals made are mostly symmetrical and normally distributed with outliers on the ends. This is expected - the chance of scoring more field goals than usual should be similar to the chance of scoring fewer than usual.



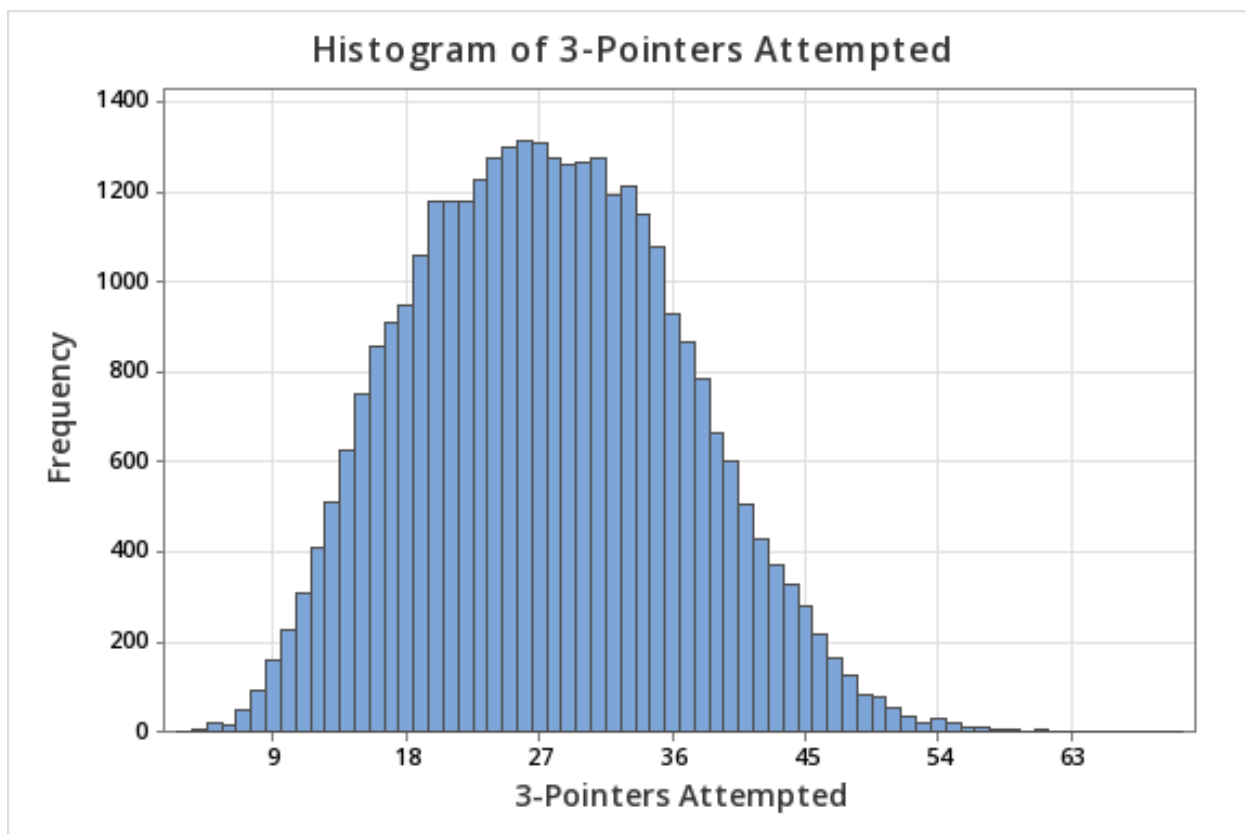
The boxplot and histogram of field goals attempted are slightly skewed to the right, with outliers on the ends and more outliers on the upper end. This makes sense when compared to field goals made, as some players can get in their heads and have poor games, causing them to miss many shots. This will increase the number of shots attempted but not affect the total number made.

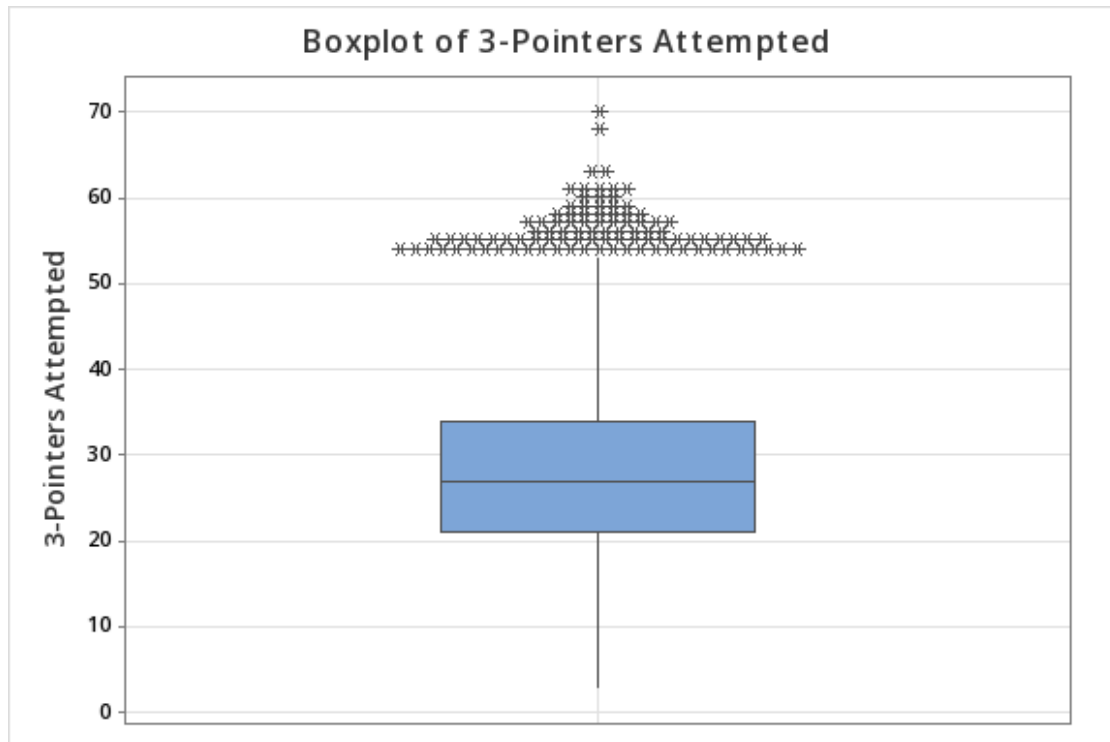


The boxplot and histogram of field goal percent are symmetrically and normally distributed with outliers on the ends. It is also interesting to notice that field goal percentage is not skewed to the right despite the skew of field goals attempted. A possible explanation for this is that NBA players are smart with their shot-taking decisions, and they will only take more field goals if they are able to maintain a good percentage of makes while doing so.

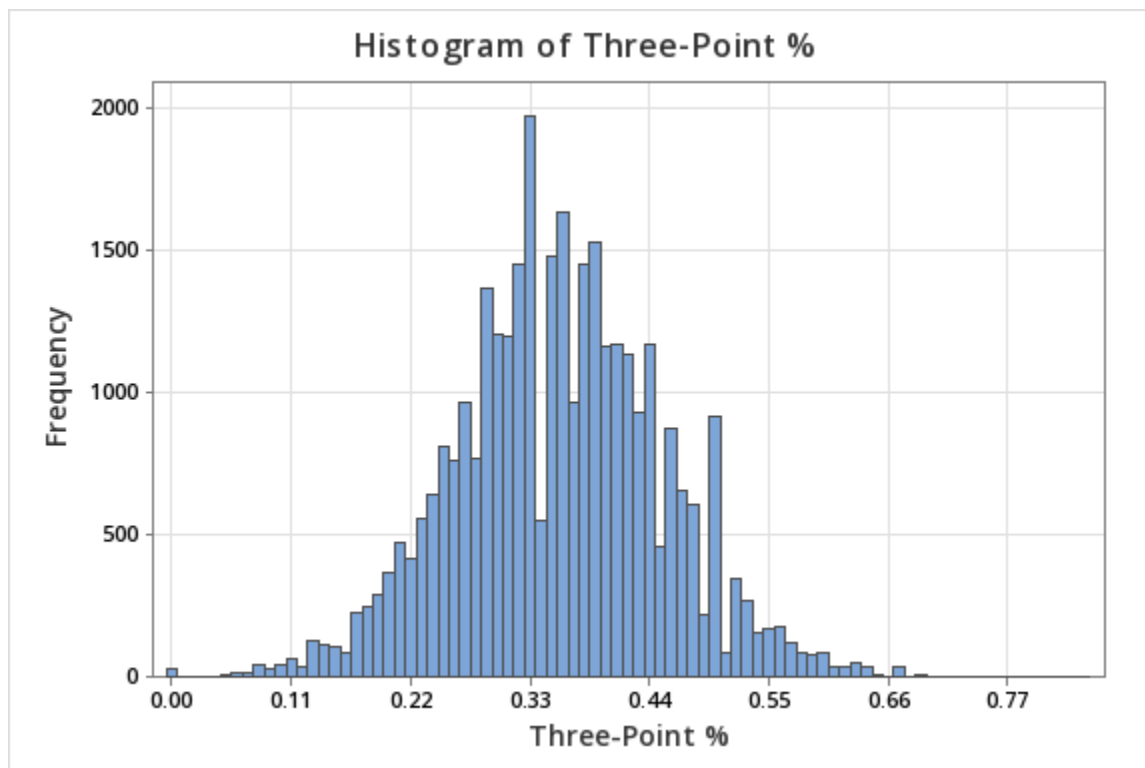


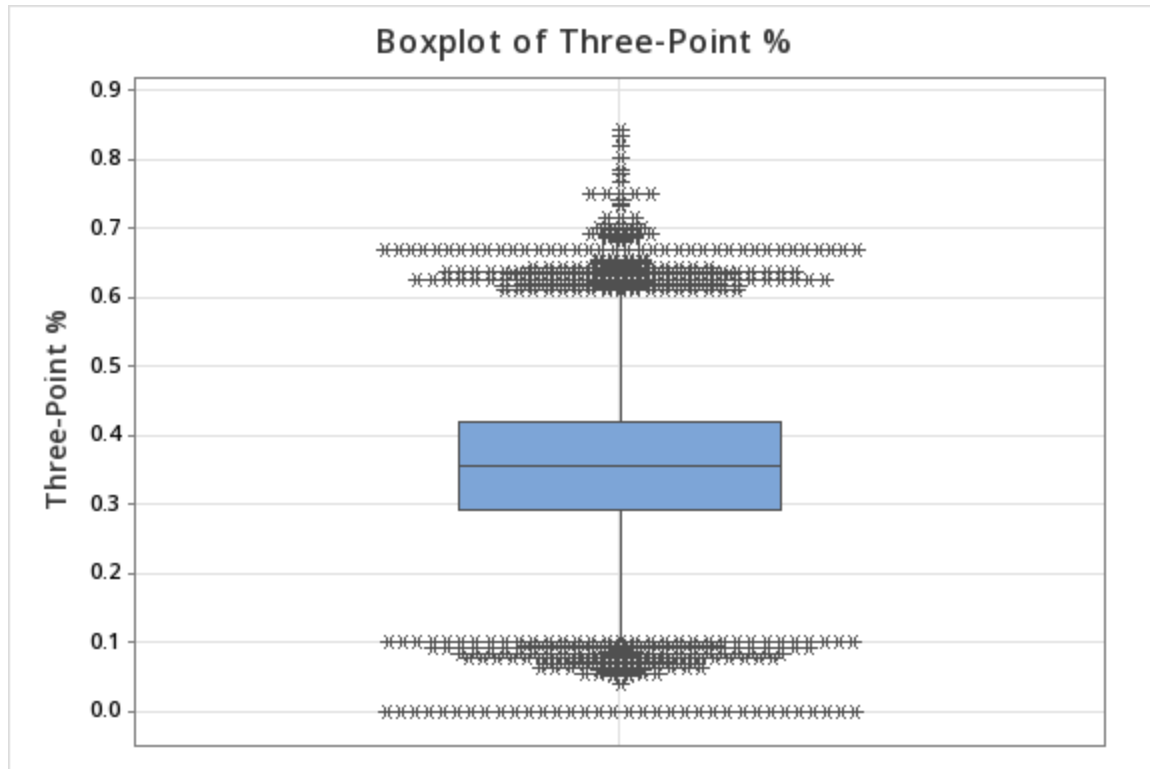
The boxplot and histogram of three pointers made are skewed to the right with outliers on the upper end. In some games, players go on what is called a “hot streak.” This is mostly mental, but it can have an incredible effect on the results of a game. After making a few shots in a row, players will sometimes continue to make shots to the point where it is almost unbelievable. In games where players do not get a hot streak they can experience the opposite effect, a cold streak. This data shows that it is much more likely to go on a hot streak than a cold streak, which makes sense because the NBA is a league of the most skilled and successful players in the world.



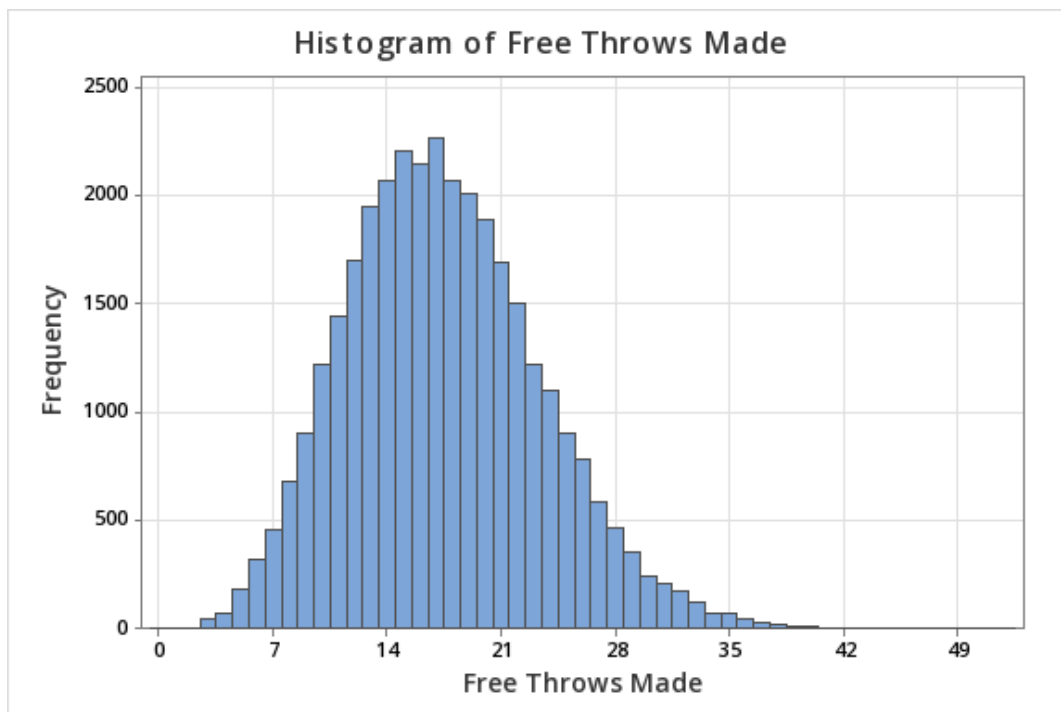


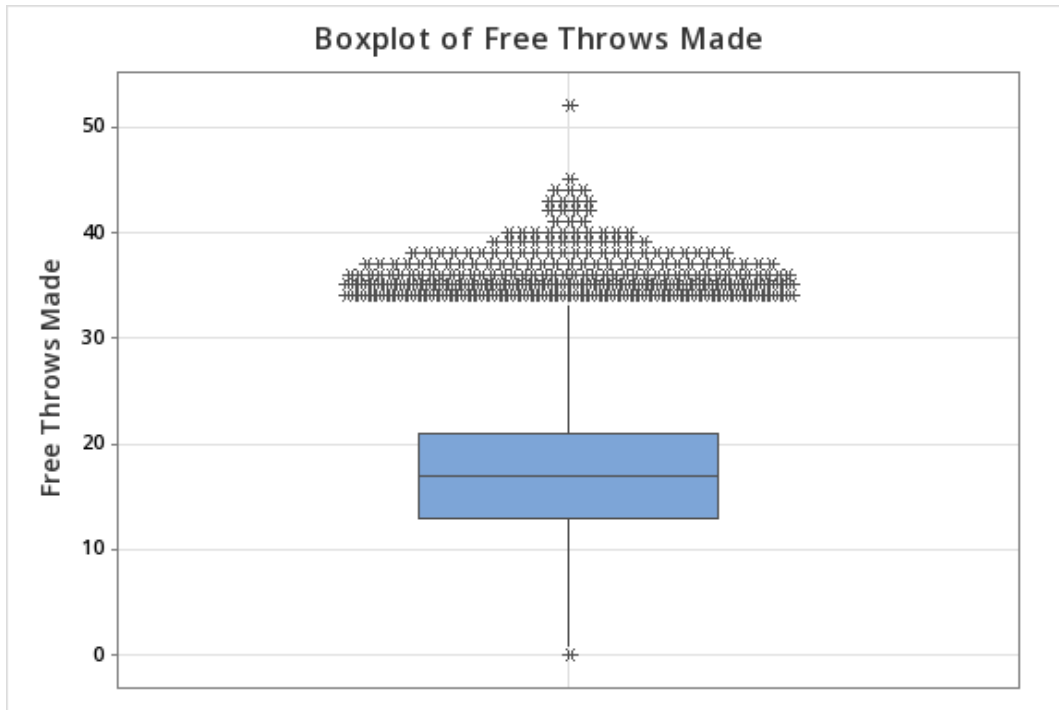
The boxplot and histogram of three pointers attempted are skewed right with outliers on the upper end. This can be explained in a similar way to three pointers made. When a player goes on a hot streak, they will be sure to take more shots because they know it will help their team win.



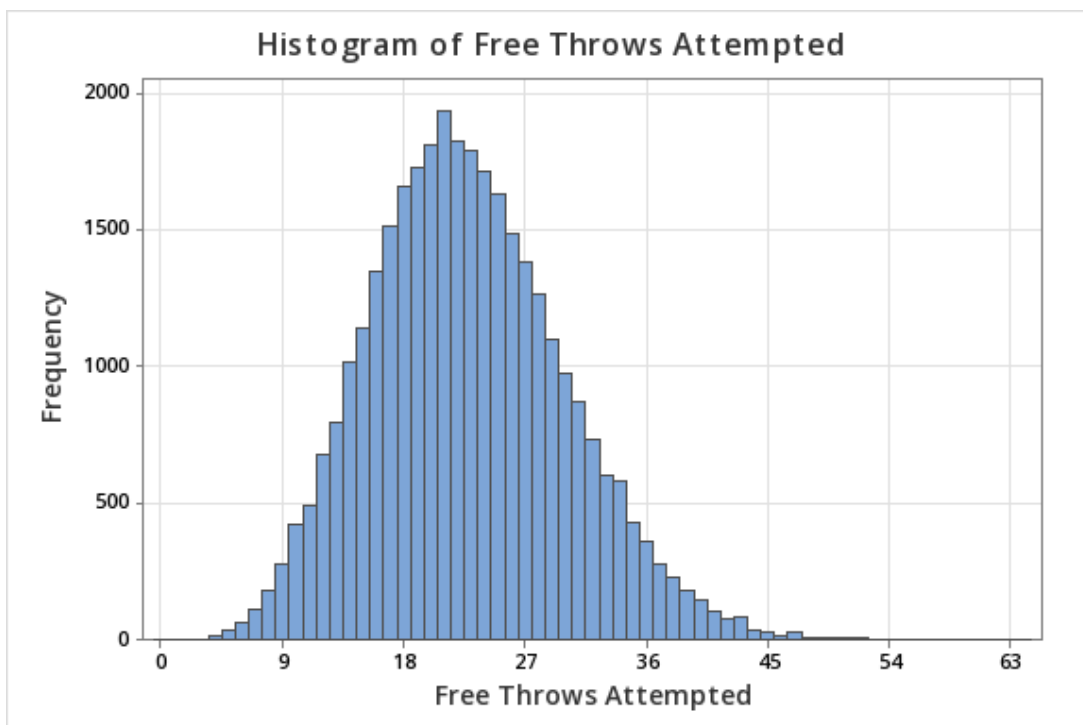


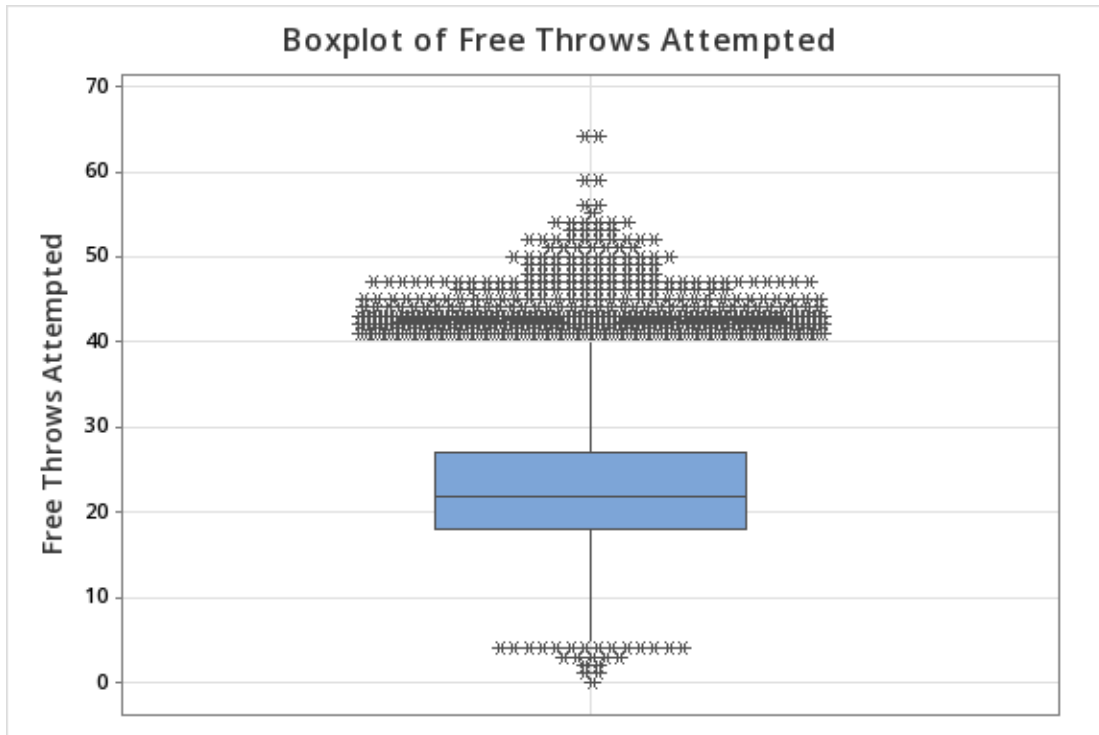
The boxplot and histogram of three point percent are somewhat normally distributed with significant outliers on the ends. This would make sense considering the hot streaks and cold streaks previously mentioned. It is a very mental game and a few shots can make or break a player's emotional state, changing the three point percentage of the team in either direction.



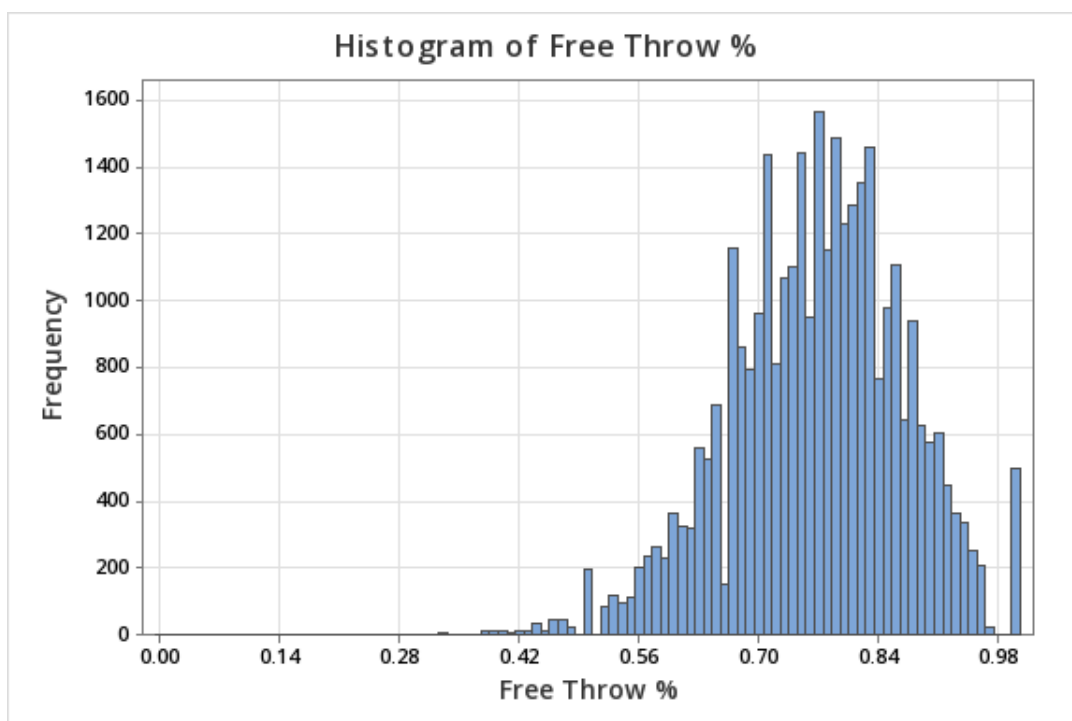


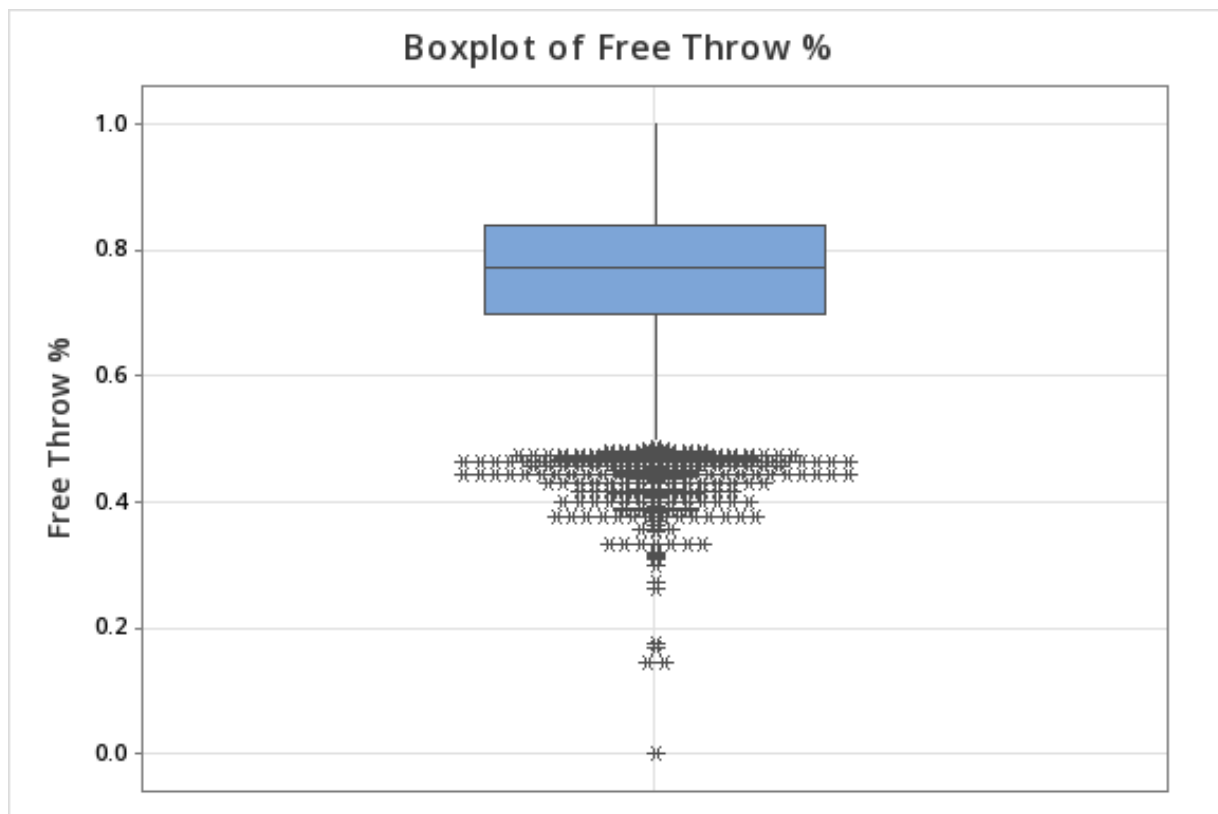
The boxplot and histogram of free throws made are skewed right with outliers on the upper end. Players can often guarantee a good number of free throws by aggressively driving and taking contact in the paint when they have the energy required. In some games, defense is unable to adapt when facing a physically dominant player, which can lead to the upper end outliers demonstrated here.



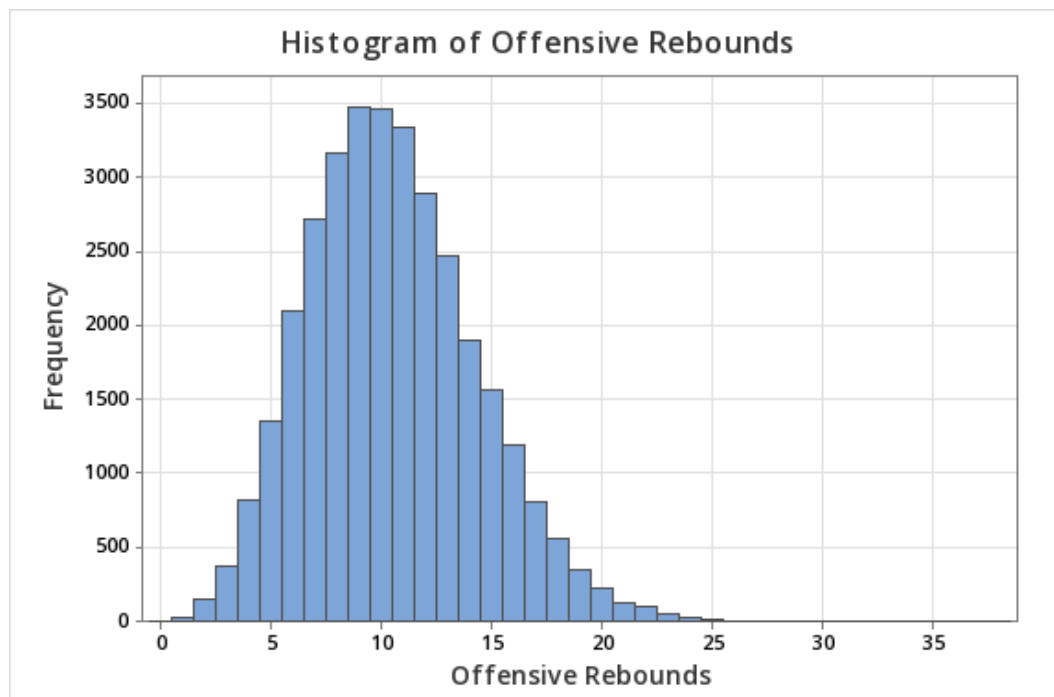


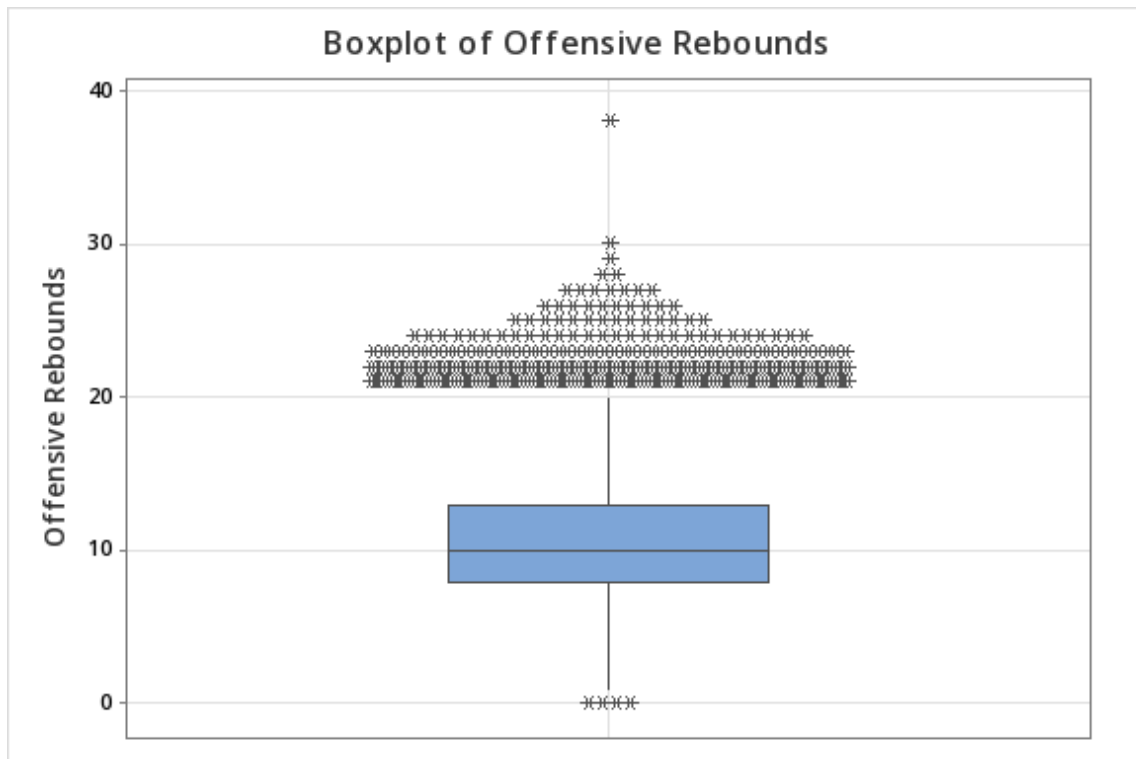
The boxplot and histogram of free throws attempted are skewed right with outliers on both ends and much more outliers on the upper end. This can be explained by the same logic as free throws made. Interestingly, because free throws are guaranteed shots that you must take if you are fouled, even if a player is a bad free throw shooter they may still attempt many free throws per game. This is not the case for field goals or three pointers.



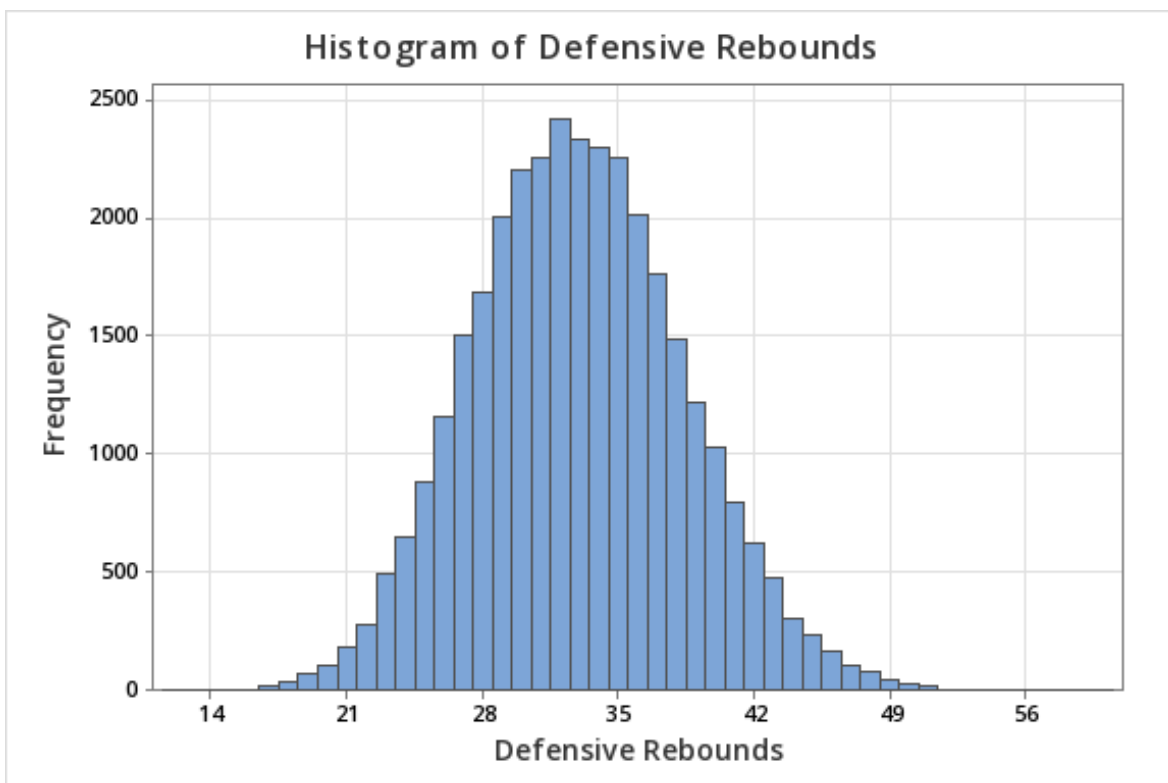


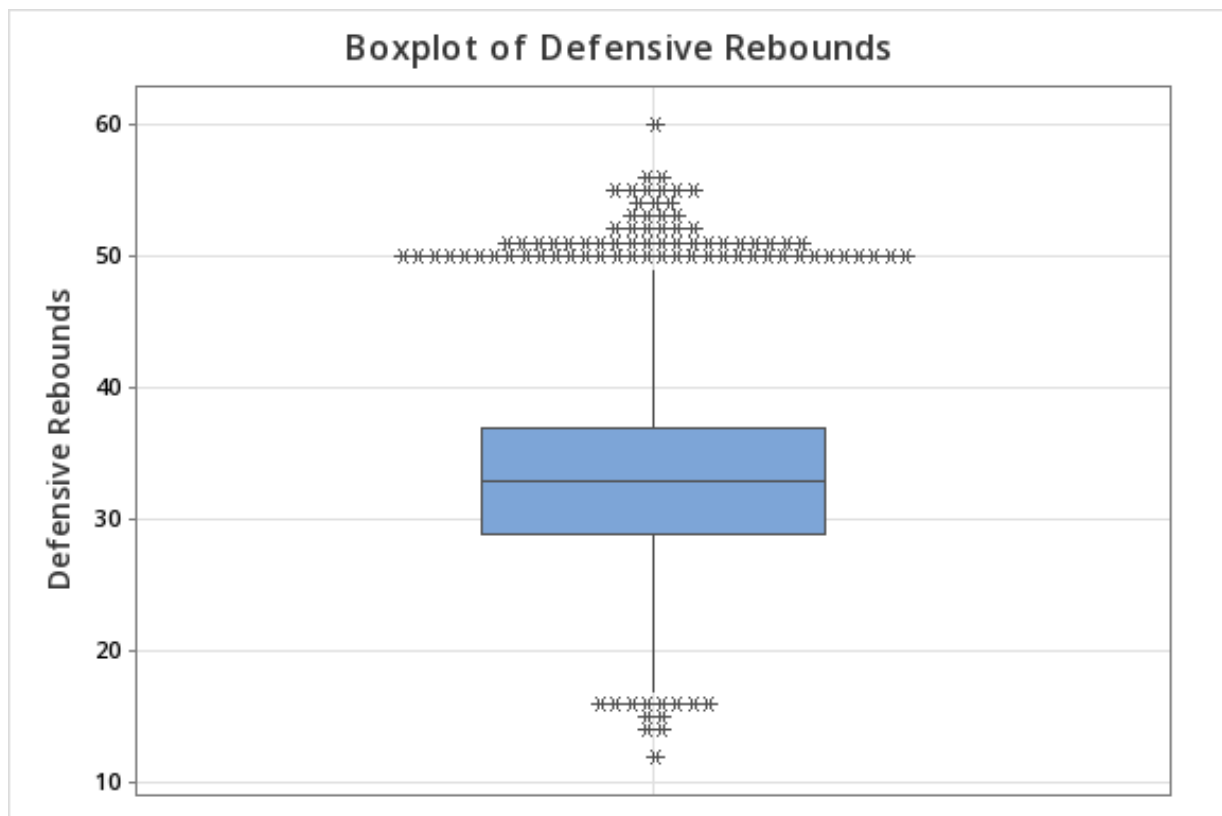
The boxplot and histogram of free throw percent are very skewed left with outliers on the lower end. This is expected, as most NBA players have great shooting ability from as close to the basket as the free throw line. However, some unique players such as those that are very tall and uncoordinated are able to play in the NBA because of their physical dominance, even without a great free throw percentage, resulting in these lower end outliers. (e.g. Shaquille O’Neal)



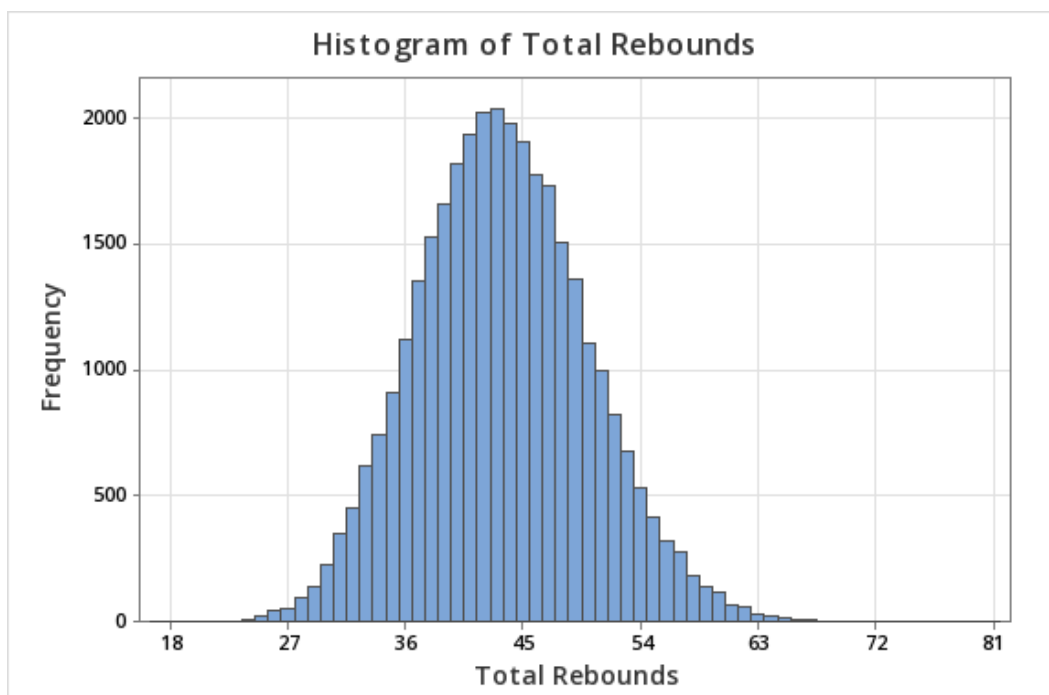


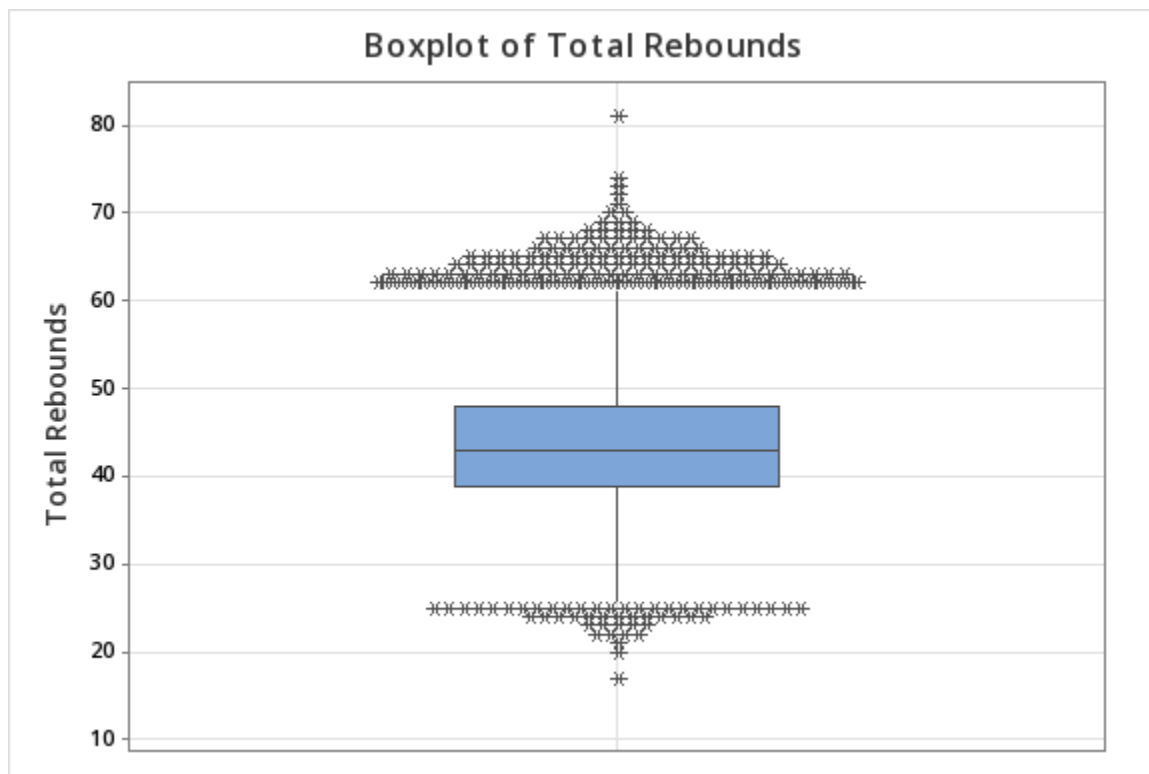
The boxplot and histogram of offensive rebounds are very skewed right with many outliers on the upper end and very few on the lower end. Once again, physically dominant players can control the game and go far beyond the baseline performance of the average NBA player.



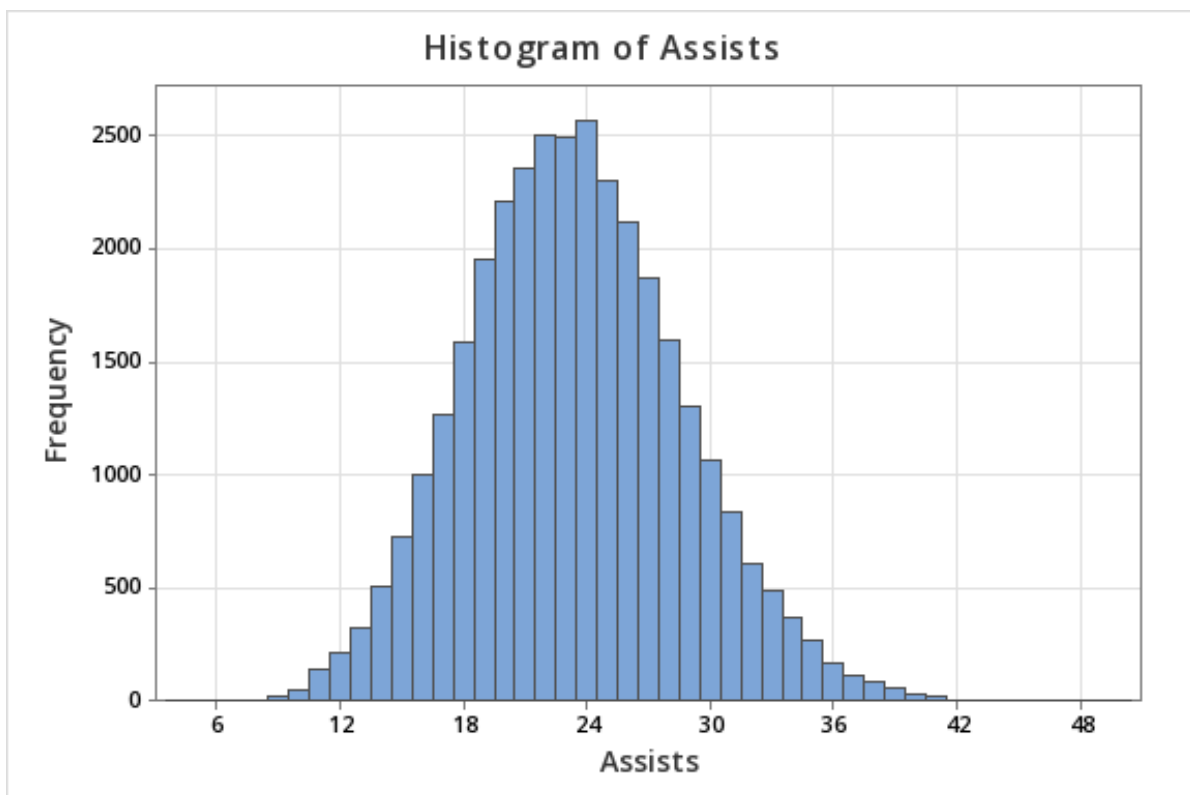


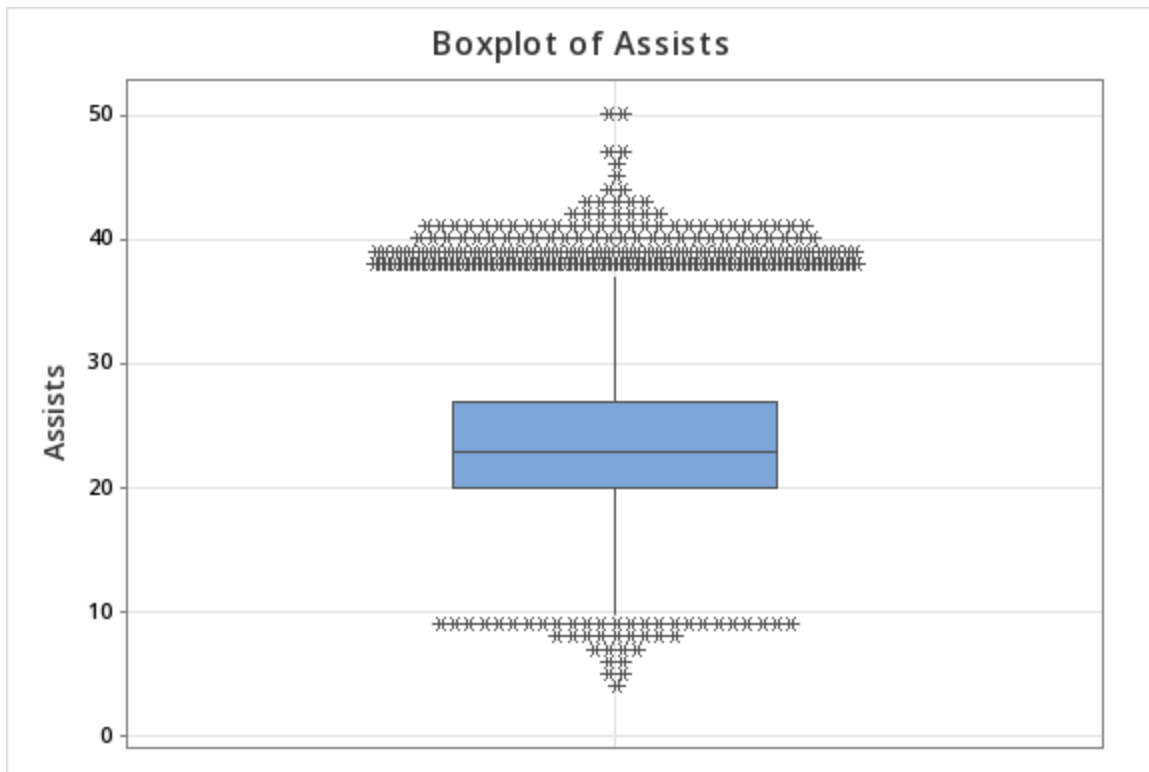
The boxplot and histogram of defensive rebounds are mostly normally distributed with outliers on both ends, and more on the upper end. An explanation for the difference between the distribution of offensive rebounds and defensive rebounds is that defensive rebounds are more under the control of the defensive team, whilst offensive rebounds are dependent on both the attacking force of the offensive team and the defensive force of the defending team.



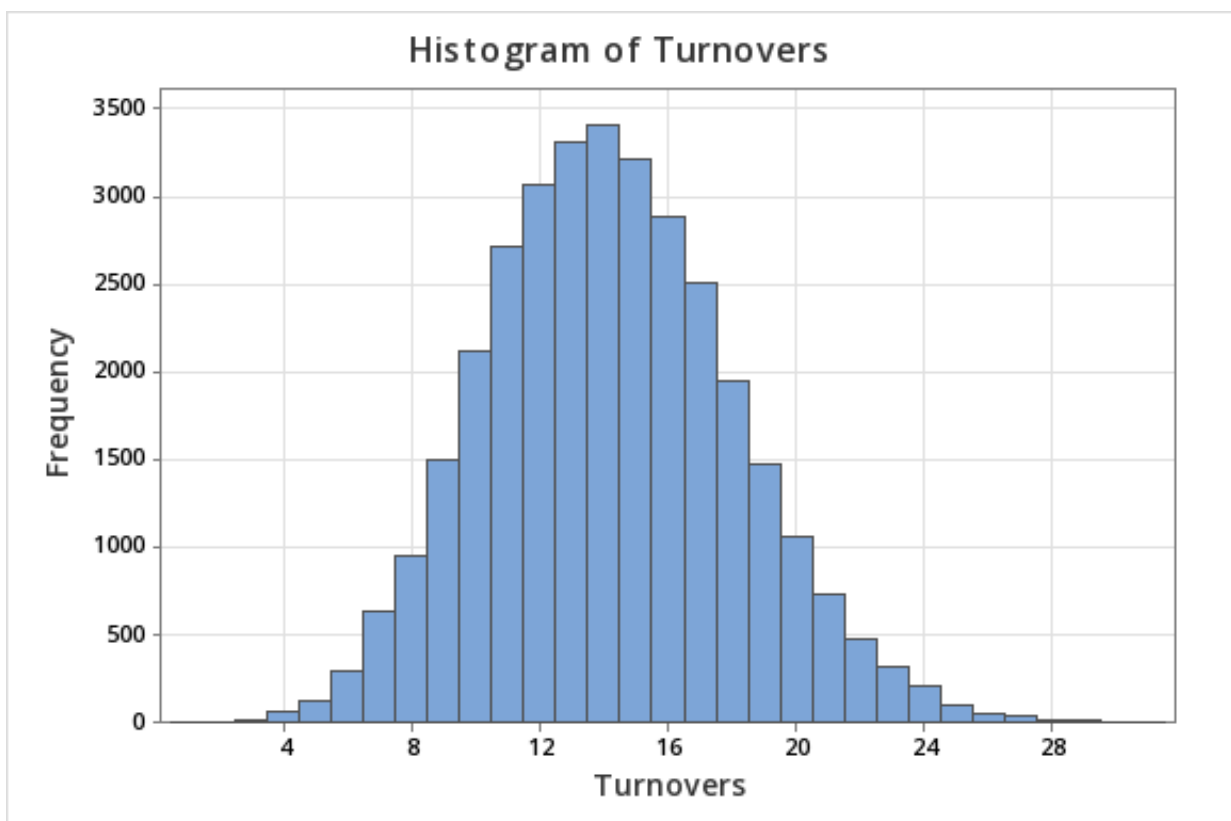


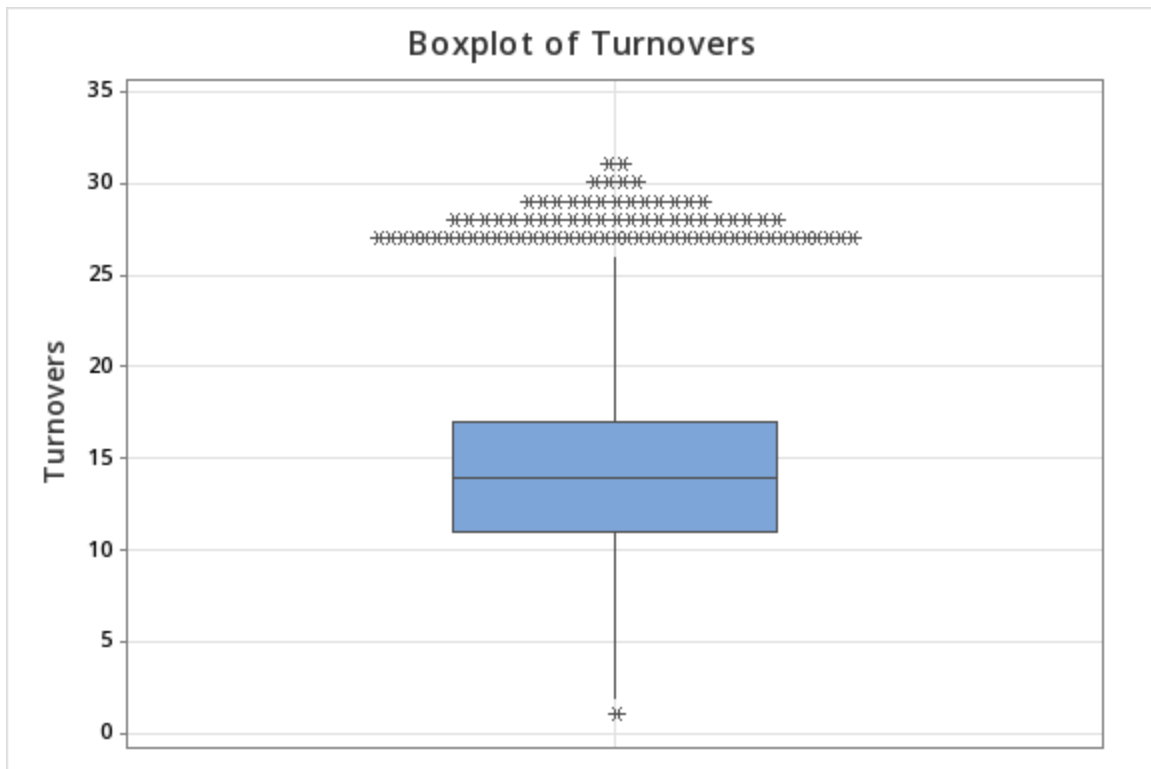
The boxplot and histogram of total rebounds are very slightly skewed right with outliers on both ends and more on the upper end. This follows logically due to the skewed distributions of both the offensive rebounds and defensive rebounds.



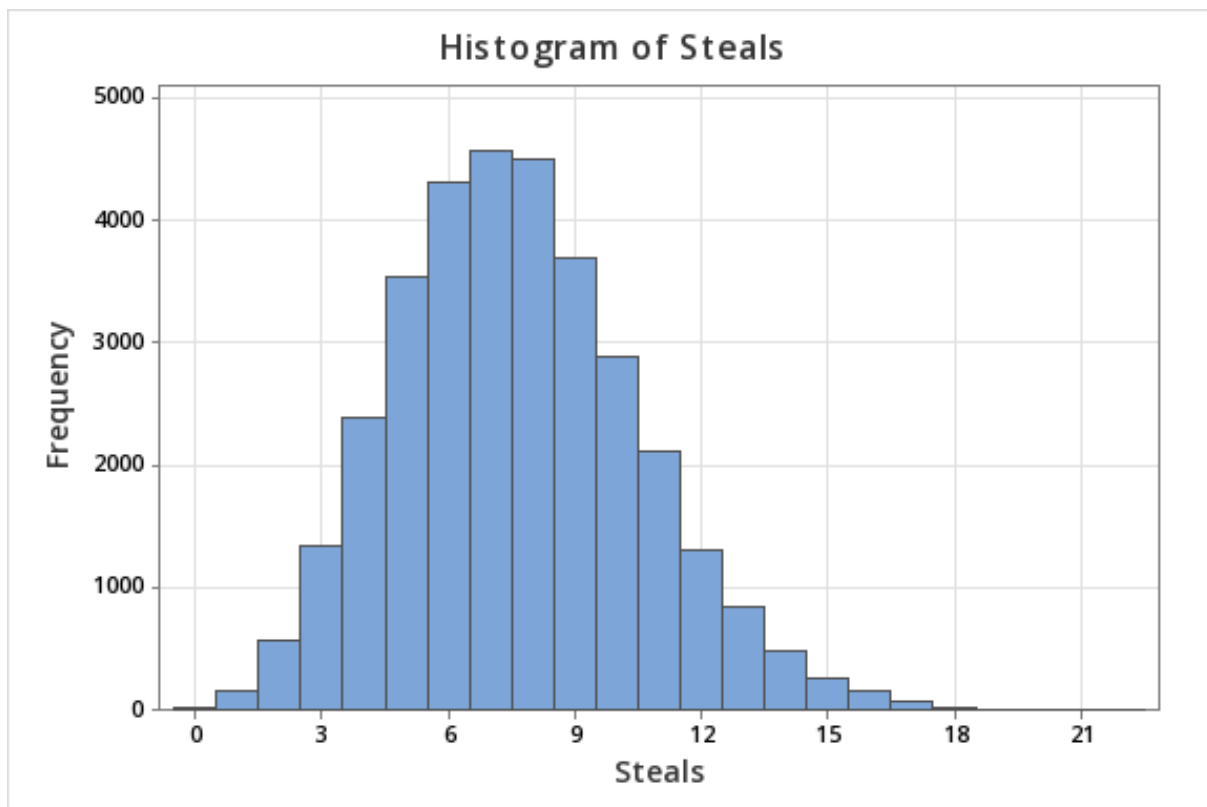


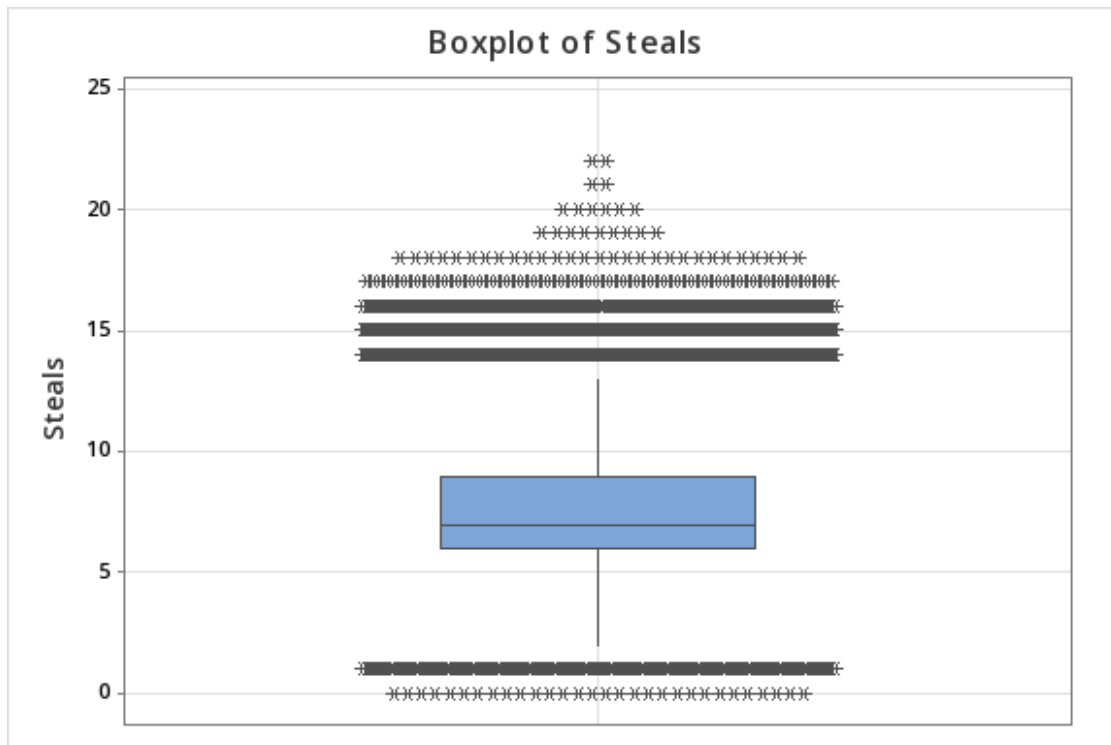
The boxplot and histogram of assists are very slightly skewed right with more outliers on the upper end and some on the lower end. A similar explanation for that of three pointers made and three pointers attempted applies here, as hot streaks also apply to assists (e.g. John Stockton).



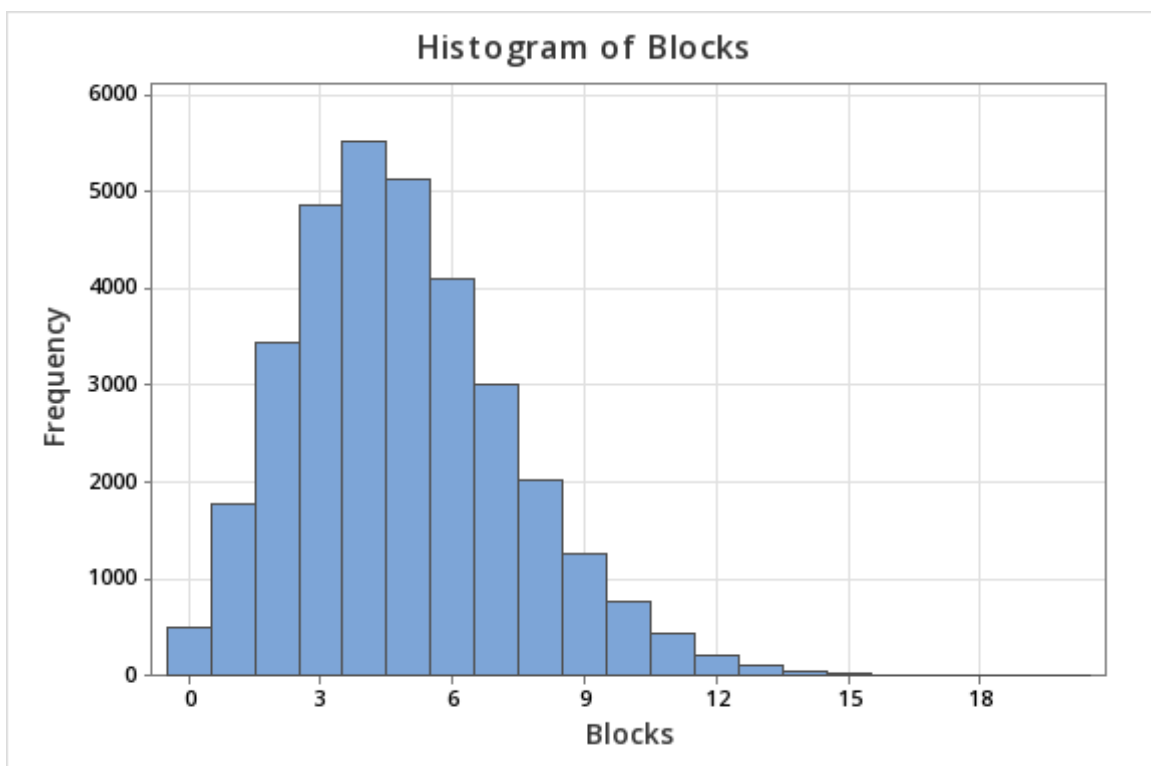


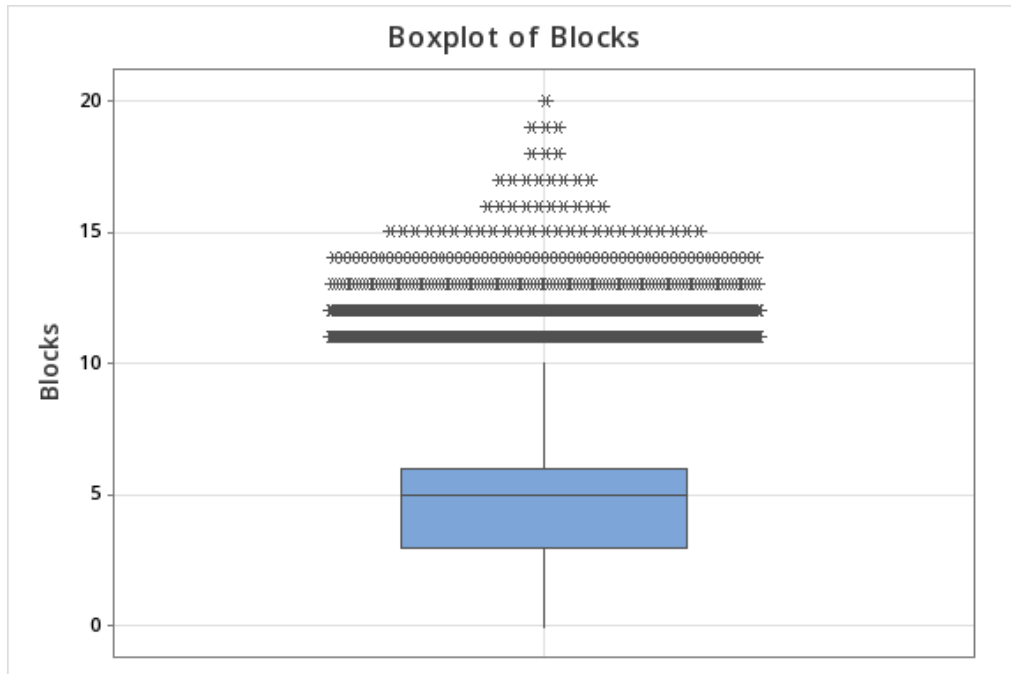
The histogram of turnovers appears to be very slightly skewed right. However, the boxplot of turnovers shows that nearly all of the outliers are on the upper end, demonstrating a distribution that is more significantly skewed to the right.



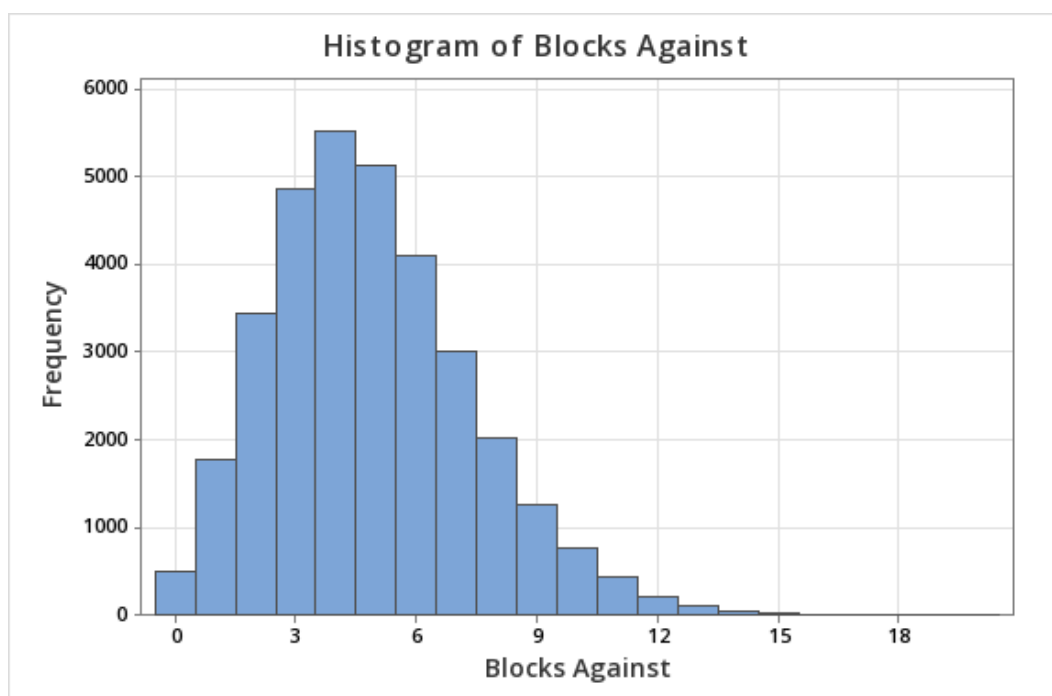


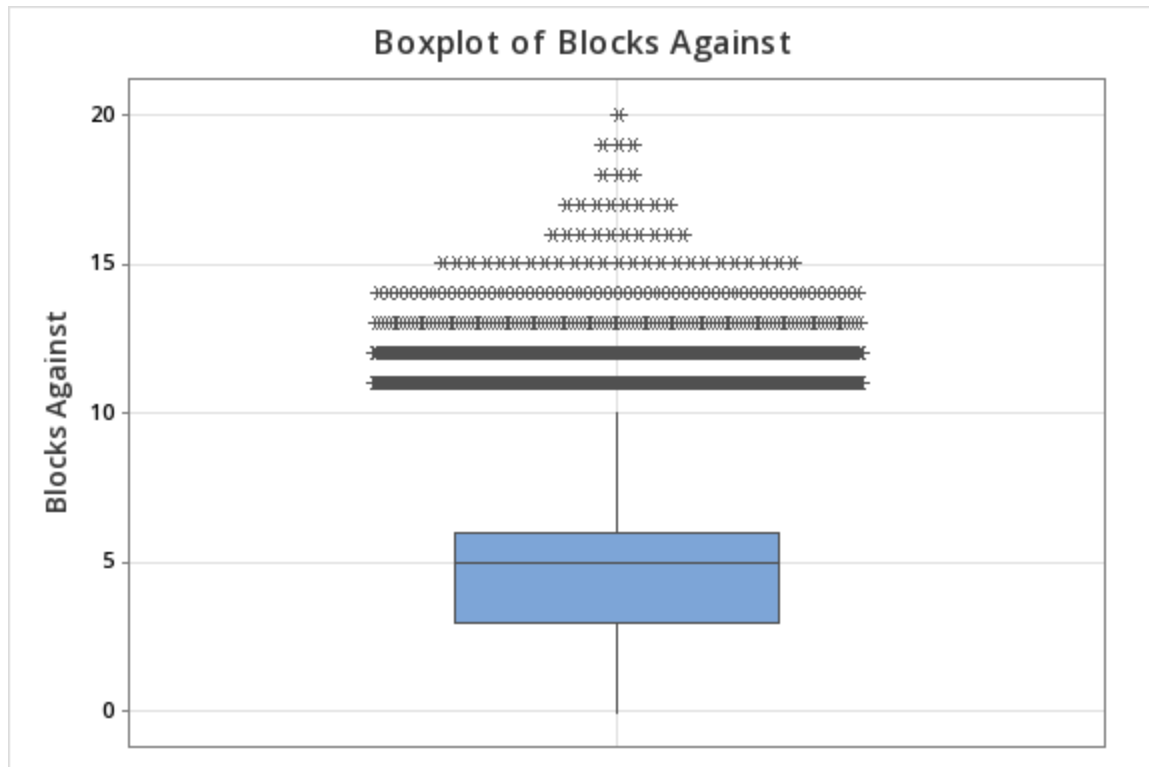
The boxplot and histogram of steals are skewed to the right with a very significant number of outliers on the upper end and still many on the lower end. This is expected, as tense games with important outcomes such as those in the playoffs are performed very carefully, without going for risky steals. However, during some regular season games, players will attempt many steals and get lucky a few times, bringing the game in their favor and resulting in an upper end outlier.



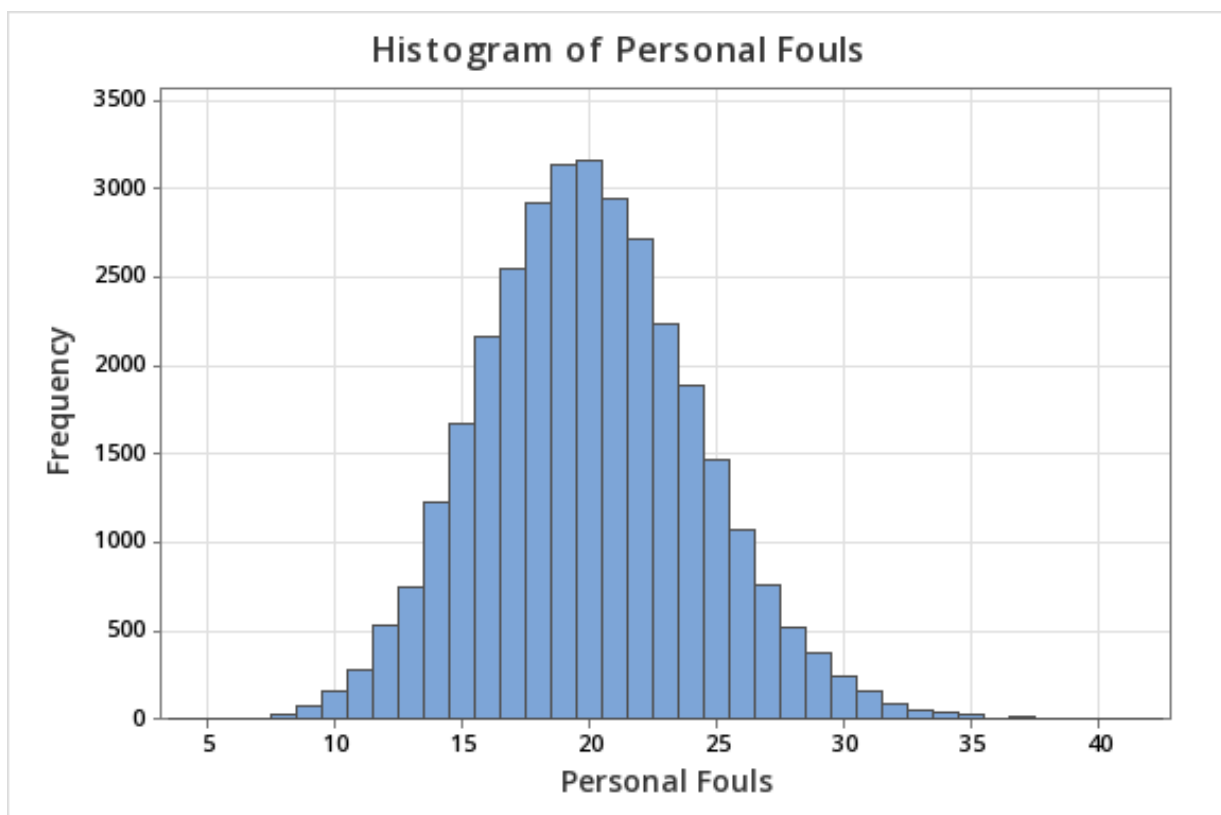


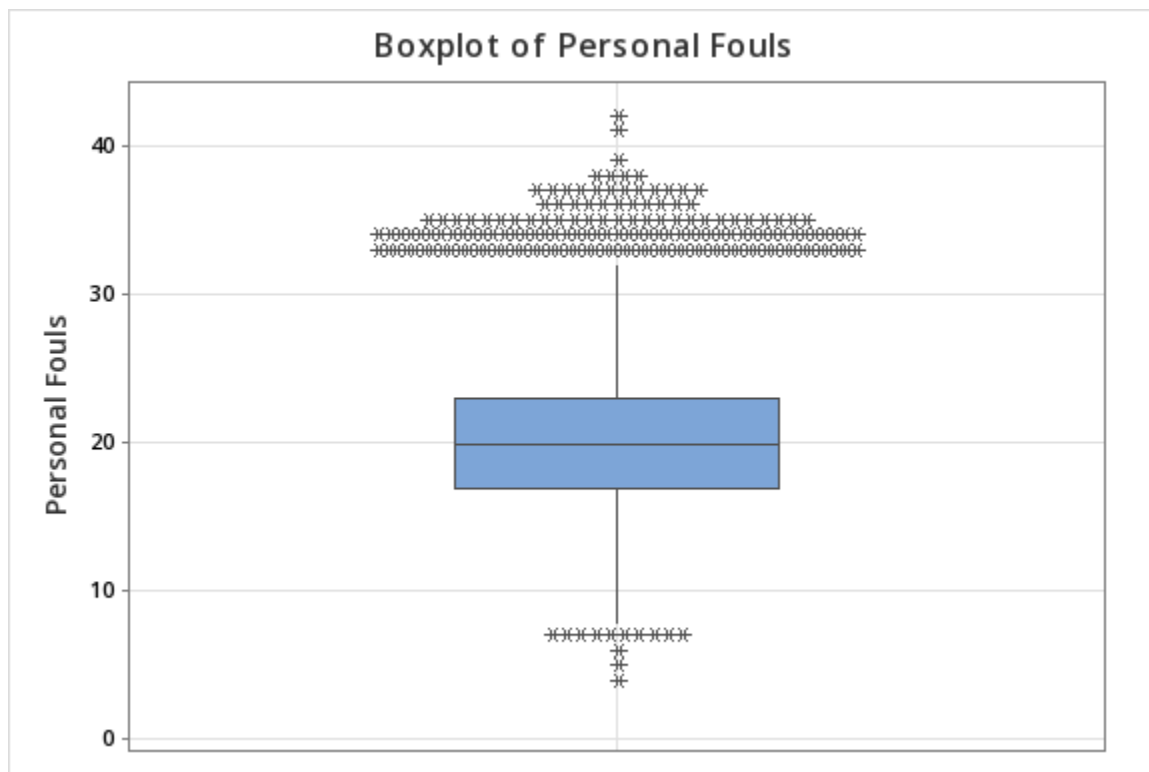
The boxplot and histogram of blocks are very skewed right with many outliers on the upper end and none on the lower end. This is the most significantly skewed distribution of all of the explanatory variables considered here. This can be explained by the idea that going for blocks is usually bad defense. Strategically, leaving your feet to attempt a block is very risk-prone to offensive strategies such as pump fakes, leading to an easy basket. Therefore, most teams would rather decrease the field goal percentage of their opponents by keeping their hands straight up and sliding their feet than try to time a block attempt perfectly.



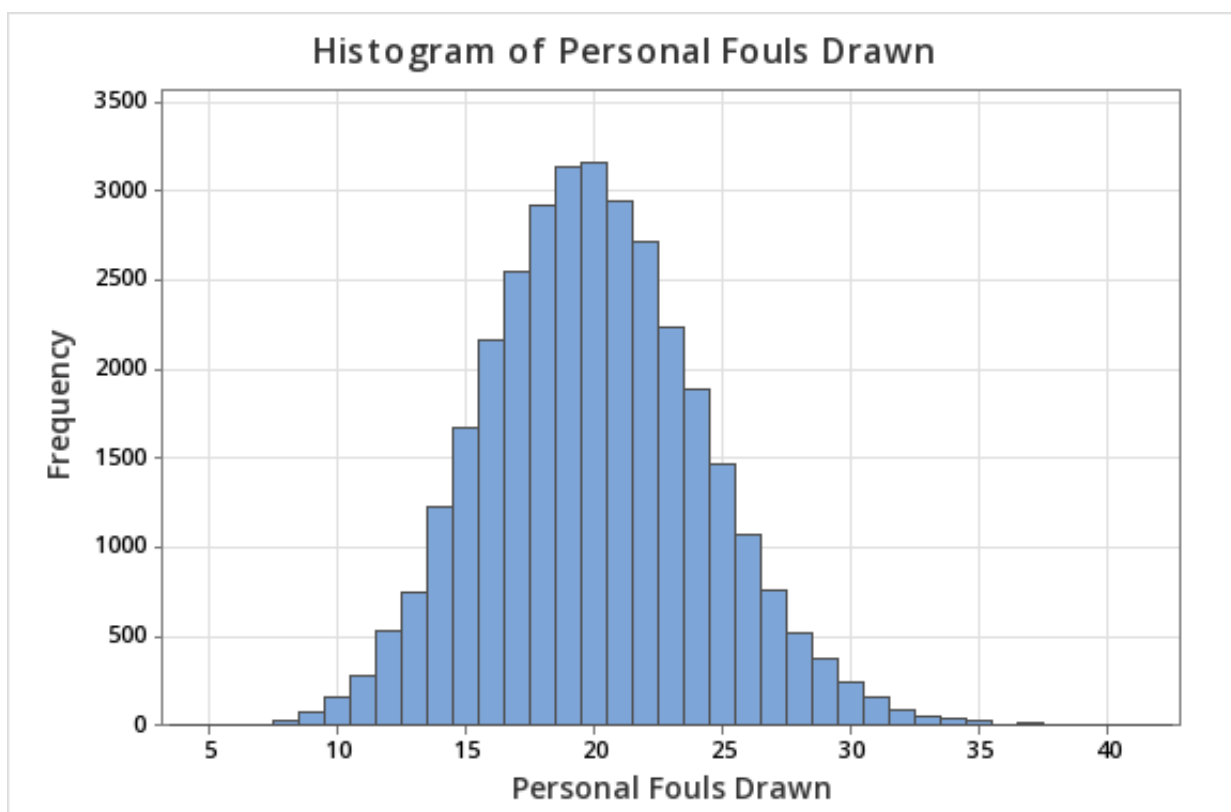


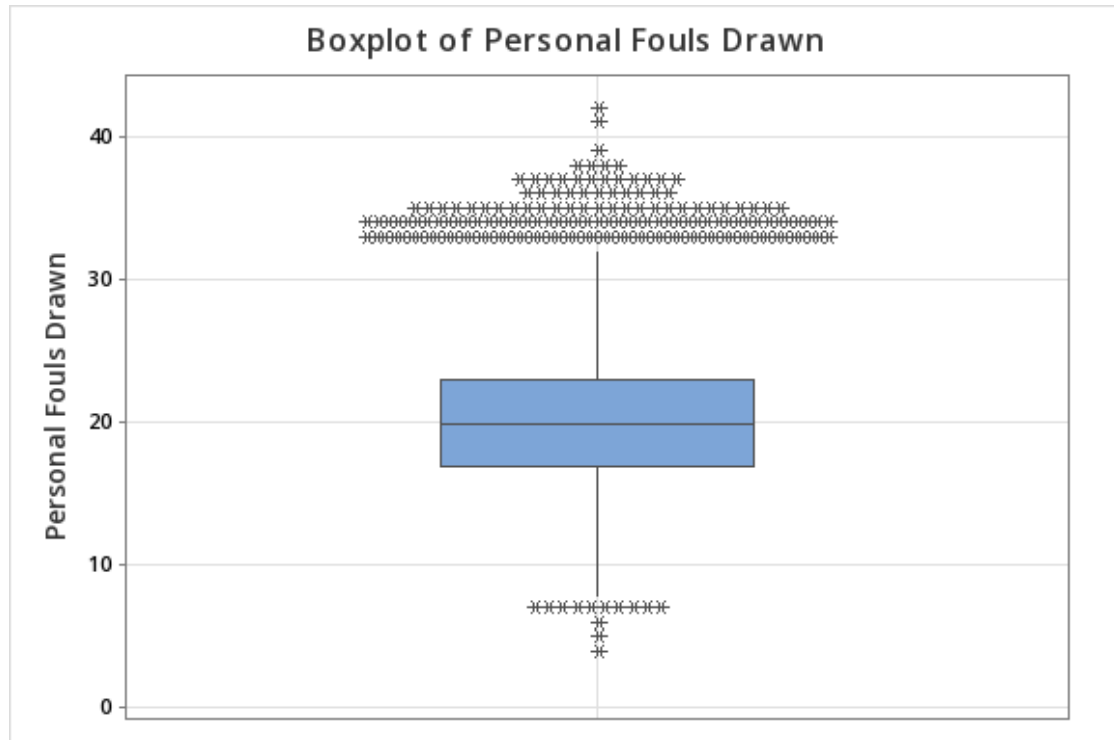
The boxplot and histogram of blocks against are very skewed right with many outliers on the upper end and none on the lower end. This can be explained with the same reasoning as for blocks. These distributions should be essentially equal, but there should be the opposite effect on point differential when included in the model.



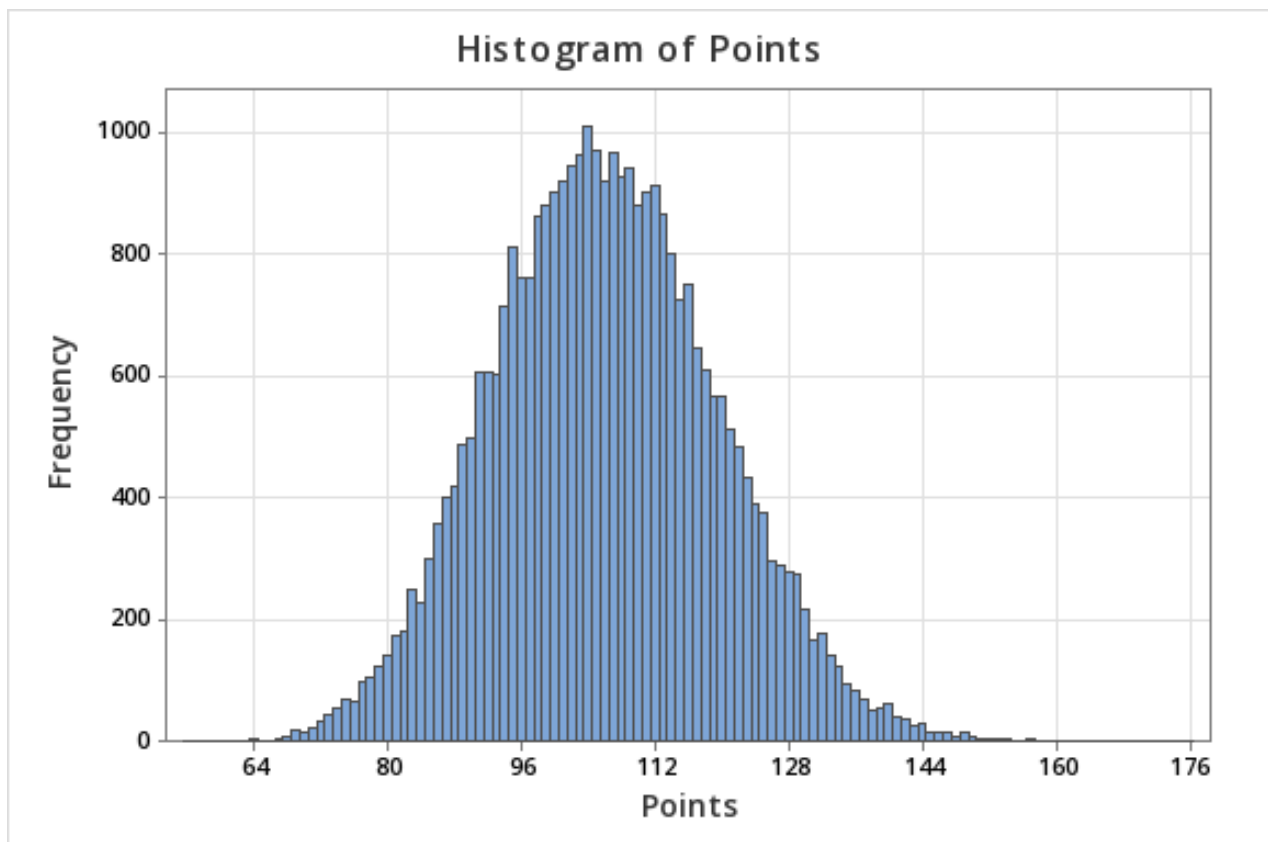


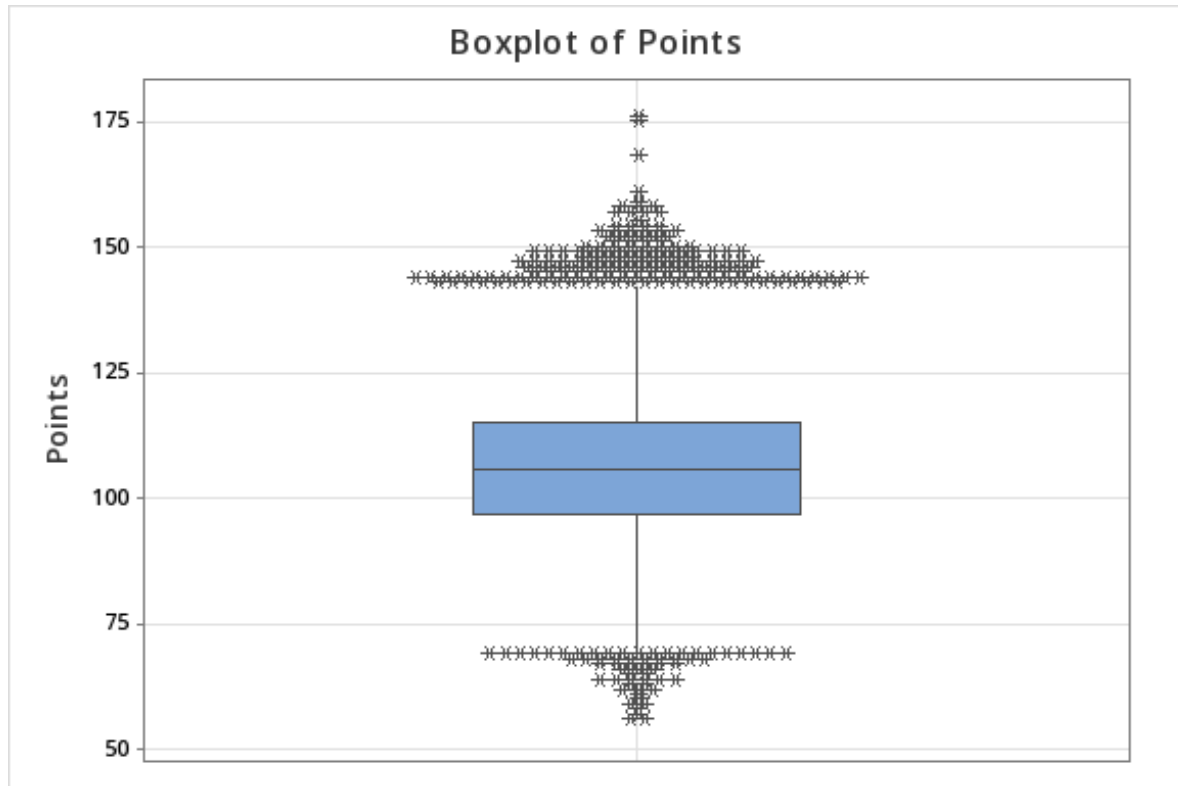
The boxplot and histogram of personal fouls are skewed right with some outliers on the upper end and few on the lower end. Most games have a baseline number of fouls which will occur out of physical exertion during the start of a match. Rarely, some NBA players push their bodies to the absolute limits when games are very important (such as the finals) leading to these outliers.





The boxplot and histogram of personal fouls drawn are skewed right with some outliers on the upper end and few on the lower end. This can be explained in the same way as personal fouls.





The boxplot and histogram of points are normally distributed with some outliers on both ends.

This can be explained by the idea that basketball is a strategic game where the only goal is to score more points than your opponent. There is no prize for scoring more points if your opponent also scores more points. There are often circumstances where a team will make offensive adjustments or defensive adjustments that work well to bring the game back to even. There is a dynamic equilibrium where both teams get small leads on each other and the team that is down always finds a way to come back. This means that points scored by a team are often left to the random chance of how well your opponents will be shooting that day and how well you will respond given the circumstances. Additionally, including this variable in the model may not be beneficial, as it is expected to have strong relationships with other explanatory variables such as field goals made and three pointers made.

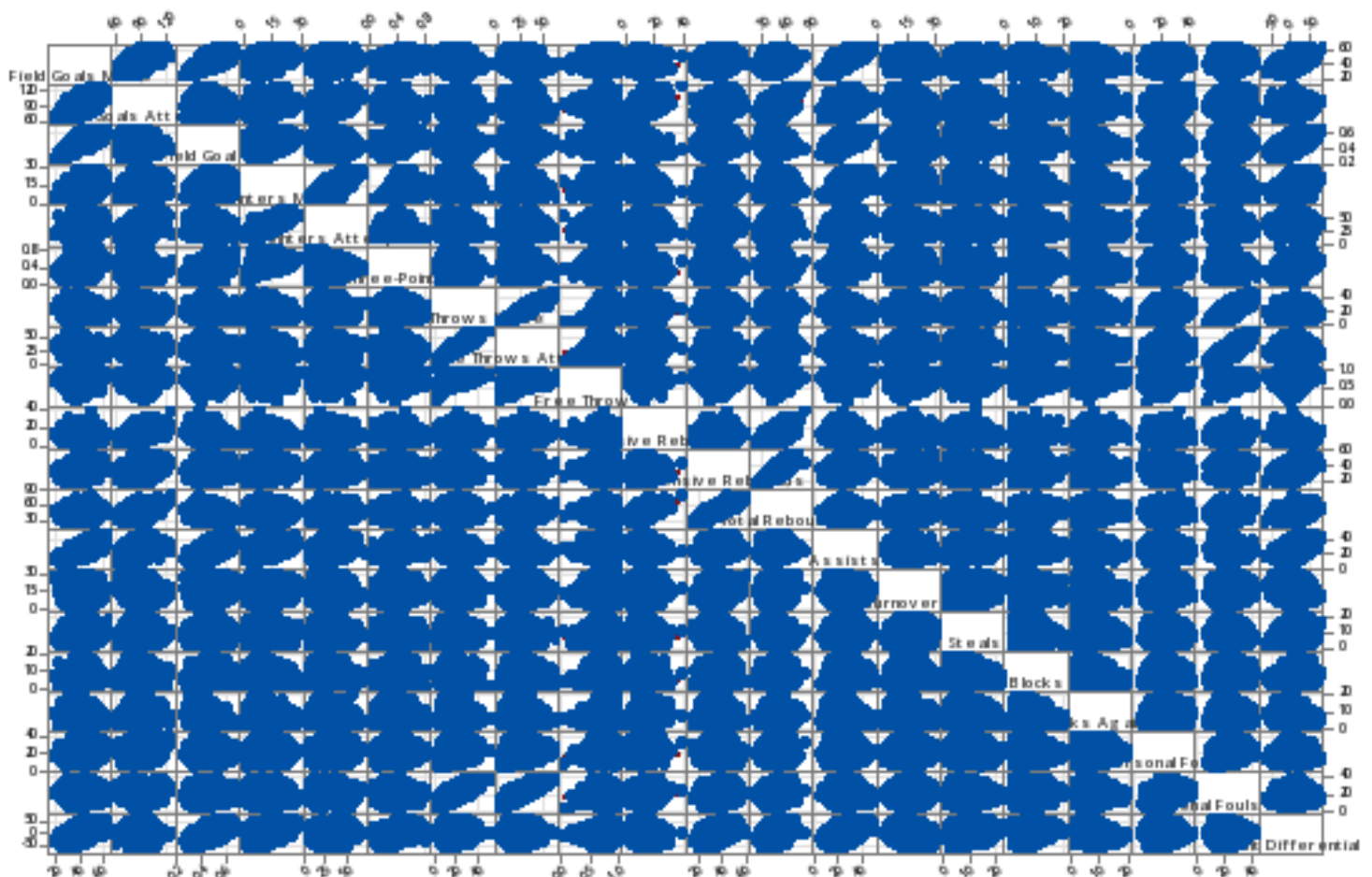
Relationship Between Pairs of Variables - Matrix Plot & Correlation Matrix:

Necessary Refinement:

When first running a regression analysis and then producing a matrix plot and correlation matrix through minitab, the existence of highly correlated predictors has been highlighted. Minitab has removed them from the analysis due to computational issues. The minitab output included this line: “The following terms cannot be estimated and were removed: Total Rebounds, Points.”

This aligns with my prior knowledge of basketball. It would make sense that as the number of total rebounds increases, the number of defensive rebounds, offensive rebounds, and field goals will also increase. This can be seen in the correlation matrix as the p-values for these correlations are very low. Additionally, points will certainly increase as field goals made, three pointers made, and free throws made increase — the low p-values also support this.

Matrix Plot of Field Goals , Field Goals , Field Goal % , ...



The large amount of data in this project makes it very difficult to read the matrix plot. This makes the pairwise Pearson correlation matrix much more useful as it demonstrates the relationships numerically through correlation coefficients. However, the matrix plot still gives me a few concerns about explanatory variables which may have high correlation with each other. These pairs include - but are not limited to - free throws attempted and free throws made, along with total rebounds and defensive rebounds. This makes me question whether we should keep these pairs of variables in our final model. We can analyze this in further detail with best-subsets.



Correlations

	Field Goals Made	Field Goals Attempted	Field Goal %	3-Pointers Made	3-Pointers Attempted	Three-Point %	Free Throws Made	Free Throws Attempted	Free Throw %	Offensive Rebounds	Defensive Rebounds	Total Rebounds	Assists	Turnovers	Steals	Blocks	Blocks Against	Personal Fouls	Personal Fouls Drawn
Field Goals Attempted	0.501																		
Field Goal %	0.764	-0.169																	
3-Pointers Made	0.432	0.260	0.296																
3-Pointers Attempted	0.264	0.426	-0.015	0.804															
Three-Point %	0.360	-0.104	0.490	0.611	0.071														
Free Throws Made	-0.161	-0.199	-0.034	-0.098	-0.079	-0.057													
Free Throws Attempted	-0.182	-0.212	-0.049	-0.128	-0.105	-0.073	0.926												
Free Throw %	0.026	-0.005	0.034	0.057	0.051	0.028	0.340	-0.015											
Offensive Rebounds	-0.006	0.466	-0.349	-0.126	-0.011	-0.195	0.044	0.085	-0.089										
Defensive Rebounds	0.163	0.197	0.039	0.153	0.187	0.017	0.045	0.046	0.009	-0.031									
Total Rebounds	0.132	0.435	-0.171	0.054	0.149	-0.099	0.063	0.088	-0.045	0.557	0.813								
Assists	0.669	0.271	0.559	0.486	0.327	0.374	-0.148	-0.169	0.027	-0.116	0.143	0.052							
Turnovers	-0.169	-0.297	0.029	-0.041	-0.066	0.018	0.020	0.031	-0.024	0.031	0.127	0.123	-0.062						
Steals	0.085	0.121	0.008	-0.010	0.010	-0.021	0.033	0.043	-0.015	0.028	-0.151	-0.109	0.073	0.094					
Blocks	0.060	0.036	0.043	-0.001	-0.012	0.011	0.016	0.018	0.001	0.019	0.184	0.164	0.068	0.046	0.009				
Blocks Against	-0.124	0.246	-0.322	-0.100	-0.065	-0.084	-0.000	0.011	-0.029	0.240	-0.020	0.123	-0.122	0.006	0.029	0.033			
Personal Fouls	0.041	0.055	0.008	0.007	0.001	0.014	0.206	0.221	0.000	0.034	-0.019	0.004	-0.019	0.151	0.028	0.011	0.010		
Personal Fouls Drawn	-0.145	-0.136	-0.062	-0.120	-0.084	-0.083	0.724	0.791	-0.032	0.074	0.004	0.046	-0.141	0.092	0.026	0.010	0.011	0.284	
Point Differential	0.488	0.010	0.547	0.290	0.053	0.390	0.097	0.057	0.120	-0.018	0.422	0.340	0.396	-0.119	0.165	0.177	-0.177	-0.039	0.039

Pairwise Pearson Correlations

Sample 1	Sample 2	N	Correlation	95% CI for p		P-Value
Field Goals Attempted	Field Goals Made	33316	0.501	(0.493,	0.509)	0.000
Field Goal %	Field Goals Made	33316	0.764	(0.760,	0.769)	0.000
3-Pointers Made	Field Goals Made	33316	0.432	(0.423,	0.440)	0.000
3-Pointers Attempted	Field Goals Made	33316	0.264	(0.254,	0.274)	0.000
Three-Point %	Field Goals Made	33316	0.360	(0.351,	0.370)	0.000
Free Throws Made	Field Goals Made	33316	-0.161	(-0.171,	-0.150)	0.000
Free Throws Attempted	Field Goals Made	33316	-0.182	(-0.193,	-0.172)	0.000
Free Throw %	Field Goals Made	33316	0.026	(0.015,	0.037)	0.000
Offensive Rebounds	Field Goals Made	33316	-0.006	(-0.017,	0.004)	0.251
Defensive Rebounds	Field Goals Made	33316	0.163	(0.152,	0.173)	0.000
Total Rebounds	Field Goals Made	33316	0.132	(0.121,	0.142)	0.000
Assists	Field Goals Made	33316	0.669	(0.663,	0.675)	0.000
Turnovers	Field Goals Made	33316	-0.169	(-0.179,	-0.158)	0.000
Steals	Field Goals Made	33316	0.085	(0.074,	0.095)	0.000
Blocks	Field Goals Made	33316	0.060	(0.049,	0.071)	0.000
Blocks Against	Field Goals Made	33316	-0.124	(-0.135,	-0.113)	0.000
Personal Fouls	Field Goals Made	33316	0.041	(0.030,	0.052)	0.000
Personal Fouls Drawn	Field Goals Made	33316	-0.145	(-0.156,	-0.135)	0.000

Sample 1	Sample 2	N	Correlation	95% CI for p		P-Value
Point Differential	Field Goals Made	33316	0.488	(0.480,	0.496)	0.000
Field Goal %	Field Goals Attempted	33316	-0.169	(-0.180,	-0.159)	0.000
3-Pointers Made	Field Goals Attempted	33316	0.260	(0.250,	0.270)	0.000
3-Pointers Attempted	Field Goals Attempted	33316	0.426	(0.417,	0.434)	0.000
Three-Point %	Field Goals Attempted	33316	-0.104	(-0.115,	-0.093)	0.000
Free Throws Made	Field Goals Attempted	33316	-0.199	(-0.209,	-0.189)	0.000
Free Throws Attempted	Field Goals Attempted	33316	-0.212	(-0.222,	-0.201)	0.000
Free Throw %	Field Goals Attempted	33316	-0.005	(-0.016,	0.005)	0.336
Offensive Rebounds	Field Goals Attempted	33316	0.466	(0.457,	0.474)	0.000
Defensive Rebounds	Field Goals Attempted	33316	0.197	(0.187,	0.208)	0.000
Total Rebounds	Field Goals Attempted	33316	0.435	(0.426,	0.444)	0.000
Assists	Field Goals Attempted	33316	0.271	(0.261,	0.281)	0.000
Turnovers	Field Goals Attempted	33316	-0.297	(-0.307,	-0.287)	0.000
Steals	Field Goals Attempted	33316	0.121	(0.110,	0.131)	0.000
Blocks	Field Goals Attempted	33316	0.036	(0.025,	0.046)	0.000
Blocks Against	Field Goals Attempted	33316	0.246	(0.236,	0.256)	0.000
Personal Fouls	Field Goals Attempted	33316	0.055	(0.045,	0.066)	0.000
Personal Fouls Drawn	Field Goals Attempted	33316	-0.136	(-0.147,	-0.125)	0.000
Point Differential	Field Goals Attempted	33316	0.010	(-0.001,	0.021)	0.068
3-Pointers Made	Field Goal %	33316	0.296	(0.286,	0.305)	0.000
3-Pointers Attempted	Field Goal %	33316	-0.015	(-0.026,	-0.005)	0.005
Three-Point %	Field Goal %	33316	0.490	(0.481,	0.498)	0.000
Free Throws Made	Field Goal %	33316	-0.034	(-0.044,	-0.023)	0.000
Free Throws Attempted	Field Goal %	33316	-0.049	(-0.060,	-0.038)	0.000
Free Throw %	Field Goal %	33316	0.034	(0.023,	0.044)	0.000
Offensive Rebounds	Field Goal %	33316	-0.349	(-0.359,	-0.340)	0.000
Defensive Rebounds	Field Goal %	33316	0.039	(0.029,	0.050)	0.000
Total Rebounds	Field Goal %	33316	-0.171	(-0.181,	-0.160)	0.000
Assists	Field Goal %	33316	0.559	(0.552,	0.566)	0.000
Turnovers	Field Goal %	33316	0.029	(0.018,	0.040)	0.000
Steals	Field Goal %	33316	0.008	(-0.003,	0.018)	0.161
Blocks	Field Goal %	33316	0.043	(0.032,	0.054)	0.000
Blocks Against	Field Goal %	33316	-0.322	(-0.331,	-0.312)	0.000
Personal Fouls	Field Goal %	33316	0.008	(-0.003,	0.018)	0.164
Personal Fouls Drawn	Field Goal %	33316	-0.062	(-0.073,	-0.051)	0.000
Point Differential	Field Goal %	33316	0.547	(0.539,	0.554)	0.000
3-Pointers Attempted	3-Pointers Made	33316	0.804	(0.800,	0.808)	0.000
Three-Point %	3-Pointers Made	33316	0.611	(0.604,	0.618)	0.000
Free Throws Made	3-Pointers Made	33316	-0.098	(-0.108,	-0.087)	0.000
Free Throws Attempted	3-Pointers Made	33316	-0.128	(-0.139,	-0.118)	0.000
Free Throw %	3-Pointers Made	33316	0.057	(0.046,	0.067)	0.000
Offensive Rebounds	3-Pointers Made	33316	-0.126	(-0.136,	-0.115)	0.000

Sample 1	Sample 2	N	Correlation	95% CI for p		P-Value
Defensive Rebounds	3-Pointers Made	33316	0.153	(0.143,	0.164)	0.000
Total Rebounds	3-Pointers Made	33316	0.054	(0.044,	0.065)	0.000
Assists	3-Pointers Made	33316	0.486	(0.478,	0.494)	0.000
Turnovers	3-Pointers Made	33316	-0.041	(-0.051,	-0.030)	0.000
Steals	3-Pointers Made	33316	-0.010	(-0.021,	0.000)	0.056
Blocks	3-Pointers Made	33316	-0.001	(-0.012,	0.010)	0.872
Blocks Against	3-Pointers Made	33316	-0.100	(-0.111,	-0.090)	0.000
Personal Fouls	3-Pointers Made	33316	0.007	(-0.004,	0.018)	0.209
Personal Fouls Drawn	3-Pointers Made	33316	-0.120	(-0.130,	-0.109)	0.000
Point Differential	3-Pointers Made	33316	0.290	(0.280,	0.299)	0.000
Three-Point %	3-Pointers Attempted	33316	0.071	(0.060,	0.081)	0.000
Free Throws Made	3-Pointers Attempted	33316	-0.079	(-0.089,	-0.068)	0.000
Free Throws Attempted	3-Pointers Attempted	33316	-0.105	(-0.116,	-0.095)	0.000
Free Throw %	3-Pointers Attempted	33316	0.051	(0.041,	0.062)	0.000
Offensive Rebounds	3-Pointers Attempted	33316	-0.011	(-0.022,	-0.001)	0.039
Defensive Rebounds	3-Pointers Attempted	33316	0.187	(0.177,	0.198)	0.000
Total Rebounds	3-Pointers Attempted	33316	0.149	(0.139,	0.160)	0.000
Assists	3-Pointers Attempted	33316	0.327	(0.317,	0.336)	0.000
Turnovers	3-Pointers Attempted	33316	-0.066	(-0.076,	-0.055)	0.000
Steals	3-Pointers Attempted	33316	0.010	(-0.001,	0.020)	0.077
Blocks	3-Pointers Attempted	33316	-0.012	(-0.023,	-0.002)	0.025
Blocks Against	3-Pointers Attempted	33316	-0.065	(-0.075,	-0.054)	0.000
Personal Fouls	3-Pointers Attempted	33316	0.001	(-0.010,	0.011)	0.897
Personal Fouls Drawn	3-Pointers Attempted	33316	-0.084	(-0.095,	-0.074)	0.000
Point Differential	3-Pointers Attempted	33316	0.053	(0.043,	0.064)	0.000
Free Throws Made	Three-Point %	33316	-0.057	(-0.067,	-0.046)	0.000
Free Throws Attempted	Three-Point %	33316	-0.073	(-0.083,	-0.062)	0.000
Free Throw %	Three-Point %	33316	0.028	(0.017,	0.039)	0.000
Offensive Rebounds	Three-Point %	33316	-0.195	(-0.205,	-0.184)	0.000
Defensive Rebounds	Three-Point %	33316	0.017	(0.007,	0.028)	0.001
Total Rebounds	Three-Point %	33316	-0.099	(-0.109,	-0.088)	0.000
Assists	Three-Point %	33316	0.374	(0.365,	0.383)	0.000
Turnovers	Three-Point %	33316	0.018	(0.007,	0.028)	0.001
Steals	Three-Point %	33316	-0.021	(-0.032,	-0.011)	0.000
Blocks	Three-Point %	33316	0.011	(0.000,	0.022)	0.042
Blocks Against	Three-Point %	33316	-0.084	(-0.095,	-0.073)	0.000
Personal Fouls	Three-Point %	33316	0.014	(0.004,	0.025)	0.009
Personal Fouls Drawn	Three-Point %	33316	-0.083	(-0.094,	-0.073)	0.000
Point Differential	Three-Point %	33316	0.390	(0.381,	0.399)	0.000
Free Throws Attempted	Free Throws Made	33316	0.926	(0.925,	0.928)	0.000
Free Throw %	Free Throws Made	33316	0.340	(0.330,	0.349)	0.000
Offensive Rebounds	Free Throws Made	33316	0.044	(0.034,	0.055)	0.000

Sample 1	Sample 2	N	Correlation	95% CI for p		P-Value
Defensive Rebounds	Free Throws Made	33316	0.045	(0.034,	0.056)	0.000
Total Rebounds	Free Throws Made	33316	0.063	(0.053,	0.074)	0.000
Assists	Free Throws Made	33316	-0.148	(-0.159,	-0.138)	0.000
Turnovers	Free Throws Made	33316	0.020	(0.009,	0.031)	0.000
Steals	Free Throws Made	33316	0.033	(0.022,	0.044)	0.000
Blocks	Free Throws Made	33316	0.016	(0.005,	0.026)	0.005
Blocks Against	Free Throws Made	33316	-0.000	(-0.011,	0.010)	0.954
Personal Fouls	Free Throws Made	33316	0.206	(0.196,	0.216)	0.000
Personal Fouls Drawn	Free Throws Made	33316	0.724	(0.719,	0.729)	0.000
Point Differential	Free Throws Made	33316	0.097	(0.087,	0.108)	0.000
Free Throw %	Free Throws Attempted	33316	-0.015	(-0.026,	-0.004)	0.007
Offensive Rebounds	Free Throws Attempted	33316	0.085	(0.075,	0.096)	0.000
Defensive Rebounds	Free Throws Attempted	33316	0.046	(0.036,	0.057)	0.000
Total Rebounds	Free Throws Attempted	33316	0.088	(0.078,	0.099)	0.000
Assists	Free Throws Attempted	33316	-0.169	(-0.179,	-0.158)	0.000
Turnovers	Free Throws Attempted	33316	0.031	(0.020,	0.042)	0.000
Steals	Free Throws Attempted	33316	0.043	(0.032,	0.053)	0.000
Blocks	Free Throws Attempted	33316	0.018	(0.007,	0.029)	0.001
Blocks Against	Free Throws Attempted	33316	0.011	(-0.000,	0.021)	0.051
Personal Fouls	Free Throws Attempted	33316	0.221	(0.210,	0.231)	0.000
Personal Fouls Drawn	Free Throws Attempted	33316	0.791	(0.787,	0.795)	0.000
Point Differential	Free Throws Attempted	33316	0.057	(0.046,	0.068)	0.000
Offensive Rebounds	Free Throw %	33316	-0.089	(-0.100,	-0.079)	0.000
Defensive Rebounds	Free Throw %	33316	0.009	(-0.002,	0.020)	0.107
Total Rebounds	Free Throw %	33316	-0.045	(-0.055,	-0.034)	0.000
Assists	Free Throw %	33316	0.027	(0.016,	0.037)	0.000
Turnovers	Free Throw %	33316	-0.024	(-0.034,	-0.013)	0.000
Steals	Free Throw %	33316	-0.015	(-0.026,	-0.004)	0.006
Blocks	Free Throw %	33316	0.001	(-0.010,	0.012)	0.848
Blocks Against	Free Throw %	33316	-0.029	(-0.040,	-0.019)	0.000
Personal Fouls	Free Throw %	33316	0.000	(-0.011,	0.011)	0.968
Personal Fouls Drawn	Free Throw %	33316	-0.032	(-0.043,	-0.021)	0.000
Point Differential	Free Throw %	33316	0.120	(0.109,	0.130)	0.000
Defensive Rebounds	Offensive Rebounds	33316	-0.031	(-0.042,	-0.020)	0.000
Total Rebounds	Offensive Rebounds	33316	0.557	(0.549,	0.564)	0.000
Assists	Offensive Rebounds	33316	-0.116	(-0.126,	-0.105)	0.000
Turnovers	Offensive Rebounds	33316	0.031	(0.020,	0.041)	0.000
Steals	Offensive Rebounds	33316	0.028	(0.018,	0.039)	0.000
Blocks	Offensive Rebounds	33316	0.019	(0.008,	0.030)	0.001
Blocks Against	Offensive Rebounds	33316	0.240	(0.230,	0.250)	0.000
Personal Fouls	Offensive Rebounds	33316	0.034	(0.023,	0.044)	0.000
Personal Fouls Drawn	Offensive Rebounds	33316	0.074	(0.064,	0.085)	0.000

Sample 1	Sample 2	N	Correlation	95% CI for p	P-Value
Point Differential	Offensive Rebounds	33316	-0.018	(-0.029, -0.007)	0.001
Total Rebounds	Defensive Rebounds	33316	0.813	(0.809, 0.817)	0.000
Assists	Defensive Rebounds	33316	0.143	(0.133, 0.154)	0.000
Turnovers	Defensive Rebounds	33316	0.127	(0.116, 0.137)	0.000
Steals	Defensive Rebounds	33316	-0.151	(-0.162, -0.141)	0.000
Blocks	Defensive Rebounds	33316	0.184	(0.174, 0.195)	0.000
Blocks Against	Defensive Rebounds	33316	-0.020	(-0.031, -0.009)	0.000
Personal Fouls	Defensive Rebounds	33316	-0.019	(-0.030, -0.008)	0.000
Personal Fouls Drawn	Defensive Rebounds	33316	0.004	(-0.007, 0.014)	0.522
Point Differential	Defensive Rebounds	33316	0.422	(0.413, 0.431)	0.000
Assists	Total Rebounds	33316	0.052	(0.041, 0.062)	0.000
Turnovers	Total Rebounds	33316	0.123	(0.113, 0.134)	0.000
Steals	Total Rebounds	33316	-0.109	(-0.120, -0.099)	0.000
Blocks	Total Rebounds	33316	0.164	(0.154, 0.175)	0.000
Blocks Against	Total Rebounds	33316	0.123	(0.113, 0.134)	0.000
Personal Fouls	Total Rebounds	33316	0.004	(-0.007, 0.014)	0.498
Personal Fouls Drawn	Total Rebounds	33316	0.046	(0.035, 0.057)	0.000
Point Differential	Total Rebounds	33316	0.340	(0.331, 0.350)	0.000
Turnovers	Assists	33316	-0.062	(-0.073, -0.051)	0.000
Steals	Assists	33316	0.073	(0.063, 0.084)	0.000
Blocks	Assists	33316	0.068	(0.057, 0.079)	0.000
Blocks Against	Assists	33316	-0.122	(-0.132, -0.111)	0.000
Personal Fouls	Assists	33316	-0.019	(-0.030, -0.008)	0.001
Personal Fouls Drawn	Assists	33316	-0.141	(-0.151, -0.130)	0.000
Point Differential	Assists	33316	0.396	(0.387, 0.405)	0.000
Steals	Turnovers	33316	0.094	(0.083, 0.104)	0.000
Blocks	Turnovers	33316	0.046	(0.035, 0.057)	0.000
Blocks Against	Turnovers	33316	0.006	(-0.004, 0.017)	0.248
Personal Fouls	Turnovers	33316	0.151	(0.141, 0.162)	0.000
Personal Fouls Drawn	Turnovers	33316	0.092	(0.081, 0.102)	0.000
Point Differential	Turnovers	33316	-0.119	(-0.130, -0.109)	0.000
Blocks	Steals	33316	0.009	(-0.002, 0.020)	0.096
Blocks Against	Steals	33316	0.029	(0.018, 0.039)	0.000
Personal Fouls	Steals	33316	0.028	(0.018, 0.039)	0.000
Personal Fouls Drawn	Steals	33316	0.026	(0.016, 0.037)	0.000
Point Differential	Steals	33316	0.165	(0.155, 0.176)	0.000
Blocks Against	Blocks	33316	0.033	(0.022, 0.043)	0.000
Personal Fouls	Blocks	33316	0.011	(0.000, 0.022)	0.043
Personal Fouls Drawn	Blocks	33316	0.010	(-0.001, 0.020)	0.076
Point Differential	Blocks	33316	0.177	(0.166, 0.187)	0.000
Personal Fouls	Blocks Against	33316	0.010	(-0.001, 0.021)	0.074
Personal Fouls Drawn	Blocks Against	33316	0.011	(0.000, 0.022)	0.042

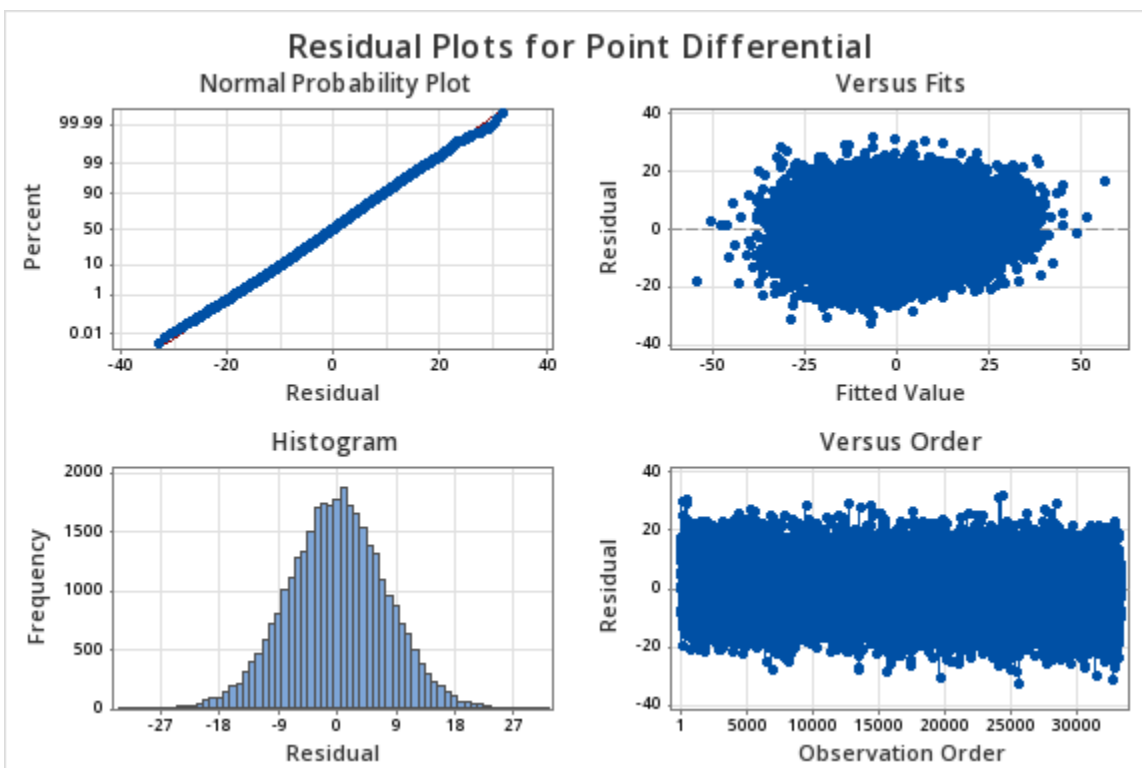
Sample 1	Sample 2	N	Correlation	95% CI for ρ	P-Value
Point Differential	Blocks Against	33316	-0.177	(-0.187, -0.166)	0.000
Personal Fouls Drawn	Personal Fouls	33316	0.284	(0.274, 0.294)	0.000
Point Differential	Personal Fouls	33316	-0.039	(-0.050, -0.028)	0.000
Point Differential	Personal Fouls Drawn	33316	0.039	(0.028, 0.050)	0.000

Analysis:

By analyzing the relationship between pairs of variables, some alarming information has become clearly evident. For one, many of the variables are highly correlated with each other, as indicated by the low p-value. Additionally, some of the explanatory variables have a small (in magnitude) correlation coefficient in their relationship with the response. The established multicollinearity of some pairs of the predictor variables is a cause for concern, as it may be difficult to determine which variables are truly significant when interpreting their individual influence. Once again, we can hope to refine this model significantly by analyzing the best-subsets of explanatory variables.

Multiple Linear Regression - Assumptions and Explanations for Inference:

Four in One Graph:



1. The observations are independent: **YES**

Each game is independent from the other games as they each occur on different dates and with differing circumstances. All observations are statistically separate and not related to any other observation.

2. The relationship between the variables is linear: **YES**

The residual plot displayed in the four in one graph demonstrates random scatter with no visible patterns. This allows us to conclude that there is most likely a linear relationship between the variables, as any other type of relationship would cause some sort of deviant pattern.

3. The standard deviation of y , σ , is the same for each value of x : **YES**

The residual plot shows that the residuals are scattered randomly with no visible patterns. There is no funnel shape or largely changing variation in the residuals over the values of x , so the overall standard deviation of the residuals is most likely to be consistent for those values of x .

4. The residuals are normally distributed: **YES**

The normal quantile plot of the residuals included in the four in one graph clearly shows a linear pattern. This tells us that the observed quantiles of the residuals and the expected quantiles (following the normal distribution) of the residuals align very well, which allows us to conclude that the residuals are indeed normally distributed.

5. The mean of the residuals is 0 : **YES**

The histogram of the residuals is mostly symmetric and centered around 0.

Conclusion Regarding Assumptions:

This data passes all 5 checks. This tells us that it is safe to perform statistical inference on the set of data using tests such as the multiple-linear regression ANOVA F test. Ultimately, passing these assumptions provides a strong foundation for drawing reliable conclusions from future analysis.

Initial Regression Information (before refinement):

Regression Equation

$$\begin{aligned} \text{Point Differential} = & -18.13 + 1.2142 \text{ Field Goals Made} - 1.2983 \text{ Field Goals Attempted} + 22.29 \text{ Field Goal \%} + 1.0782 \\ & 3\text{-Pointers Made} - 0.2789 \text{ 3-Pointers Attempted} - 1.66 \text{ Three-Point \%} + 0.6629 \text{ Free Throws} \\ & \text{Made} - 0.8685 \text{ Free Throws Attempted} + 3.17 \text{ Free Throw \%} + 1.5217 \text{ Offensive Rebounds} + \\ & 1.45805 \text{ Defensive Rebounds} + 0.1135 \text{ Assists} - 1.3649 \text{ Turnovers} + 1.5888 \text{ Steals} + 0.3883 \\ & \text{Blocks} + 0.0708 \text{ Blocks Against} + 0.0364 \text{ Personal Fouls} + 0.6149 \text{ Personal Fouls Drawn} \end{aligned}$$

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-18.13	4.00	-4.53	0.000	
Field Goals Made	1.2142	0.0998	12.17	0.000	168.25
Field Goals Attempted	-1.2983	0.0468	-27.72	0.000	74.51
Field Goal %	22.29	8.46	2.63	0.008	128.12
3-Pointers Made	1.0782	0.0487	22.14	0.000	24.86
3-Pointers Attempted	-0.2789	0.0183	-15.22	0.000	16.29
Three-Point %	-1.66	1.23	-1.34	0.179	8.37
Free Throws Made	0.6629	0.0541	12.24	0.000	60.94
Free Throws Attempted	-0.8685	0.0428	-20.30	0.000	56.71
Free Throw %	3.17	1.18	2.68	0.007	8.62
Offensive Rebounds	1.5217	0.0146	104.03	0.000	1.83
Defensive Rebounds	1.45805	0.00868	168.06	0.000	1.31
Assists	0.1135	0.0113	10.06	0.000	2.05
Turnovers	-1.3649	0.0128	-106.73	0.000	1.44
Steals	1.5888	0.0153	103.86	0.000	1.14
Blocks	0.3883	0.0168	23.16	0.000	1.05
Blocks Against	0.0708	0.0184	3.85	0.000	1.26
Personal Fouls	0.0364	0.0104	3.48	0.001	1.17
Personal Fouls Drawn	0.6149	0.0164	37.58	0.000	2.86

The multiple linear regression equation is able to predict the point differential of a game based on 18 predictor variables. The coefficient (Coef) of each of the predictor variables represents the change in point differential for every increase by one unit in the specific predictor, assuming that the other predictors are held constant. For example, for every increment in free throws made, the point differential is expected to increase by 0.6629 points when all the other predictors are fixed. For every increase by one in turnovers, point differential is expected to decrease by 1.3649 points when all other predictors are fixed. These slopes definitely align with my understanding of basketball, as turning over the ball will lose your team a possession and give the other team a chance to score, and scoring a field goal will immediately increase your team's score.

Analysis of Variance					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
☒ Regression	18	4774450	265247	4610.22	0.000
Field Goals Made	1	8517	8517	148.04	0.000
Field Goals Attempted	1	44213	44213	768.46	0.000
Field Goal %	1	399	399	6.94	0.008
3-Pointers Made	1	28201	28201	490.16	0.000
3-Pointers Attempted	1	13329	13329	231.68	0.000
Three-Point %	1	104	104	1.80	0.179
Free Throws Made	1	8624	8624	149.89	0.000
Free Throws Attempted	1	23703	23703	411.98	0.000
Free Throw %	1	415	415	7.21	0.007
Offensive Rebounds	1	622666	622666	10822.45	0.000
Defensive Rebounds	1	1625003	1625003	28243.89	0.000
Assists	1	5822	5822	101.19	0.000
Turnovers	1	655419	655419	11391.73	0.000
Steals	1	620570	620570	10786.01	0.000
Blocks	1	30865	30865	536.46	0.000
Blocks Against	1	854	854	14.84	0.000
Personal Fouls	1	696	696	12.11	0.001
Personal Fouls Drawn	1	81242	81242	1412.05	0.000
Error	33297	1915732	58		
Total	33315	6690182			

Model Summary			
S	R-sq	R-sq(adj)	R-sq(pred)
7.58516	71.37%	71.35%	71.33%

Further Analysis:

A very good sign for this data is that every explanatory variable except for three point percentage is very correlated with point differential. Each of these explanatory variables has a very low p-value, indicating that the linear relationship is very significant. For each of these explanatory variables, we can conclude that it does influence point differential when the other explanatory variables are included in the model. Additionally, it is important to consider removing three point percentage from the final model as it is by far the weakest predictor. From a basketball standpoint, this relationship is not surprising as a team may shoot only one or two three pointers in an entire game but make them all, yielding a three point percentage of 100%. Another thing to consider from this information is the moderately strong value of R-squared. 71% is not optimal, so once again we can hope to further refine this model through best-subsets.

Best Subsets - Refining the Regression Model:

Response is Point Differential

	R-Sq (adj)	R-Sq (pred)	Mallows Cp	S	Field Goals Made	Field Goals Attempted	Field Goal %	3- Pointers Made	3-Pointers Attempted	Three- Point %	Free Throws Made	Free Throws Attempted	Free Throw %	Offensive Rebounds	Defensive Rebounds	Assists	Turnovers	Steals	Blocks	Blocks Against	Personal Fouls	Personal Fouls Drawn
1	29.9	29.9	29.9	48193.4	11.864		X															
1	23.8	23.8	23.8	55301.7	12.371	X																
2	46.0	46.0	46.0	29527.2	10.418			X							X							
2	35.8	35.8	35.8	41296.3	11.351	X									X							
3	51.0	51.0	51.0	23657.7	9.9191		X								X			X				
3	49.6	49.6	49.6	25243.1	10.056		X						X		X							
4	55.6	55.6	55.6	18317.5	9.4427		X								X		X	X				
4	54.4	54.4	54.4	19672.8	9.5658		X							X	X			X				
5	59.4	59.4	59.4	13916.9	9.0312		X							X	X		X	X				
5	57.8	57.8	57.8	15733.0	9.2032		X			X					X		X	X				
6	64.8	64.8	64.8	7652.1	8.4107	X	X							X	X		X	X				
6	64.5	64.5	64.5	7927.6	8.4389	X		X						X	X		X	X				
7	67.1	67.1	67.1	4903.0	8.1234	X	X			X				X	X		X	X				
7	67.1	67.1	67.1	4923.4	8.1256	X	X		X					X	X		X	X				
8	68.6	68.6	68.6	3238.6	7.9444	X	X			X				X	X		X	X				
8	68.5	68.5	68.5	3352.5	7.9568	X	X		X				X	X	X		X	X				
9	69.3	69.3	69.3	2393.2	7.8519	X	X		X				X	X	X		X	X				
9	69.2	69.2	69.2	2533.1	7.8672		X	X	X				X	X	X		X	X				
10	70.0	70.0	70.0	1590.6	7.7630	X	X		X		X	X		X	X		X	X				X
10	69.9	69.9	69.9	1670.3	7.7718	X	X			X	X	X		X	X		X	X				X
11	70.8	70.7	70.7	716.8	7.6650	X	X		X	X		X		X	X		X	X				X
11	70.6	70.6	70.6	852.7	7.6803	X	X		X	X		X	X	X	X		X	X				X
12	71.2	71.2	71.2	147.6	7.6005	X	X		X		X	X		X	X		X	X	X			X
12	71.1	71.1	71.1	289.5	7.6166	X	X		X	X		X	X	X	X		X	X	X			X
13	71.3	71.3	71.3	49.8	7.5892	X	X		X	X		X	X		X	X	X	X	X			X
13	71.3	71.2	71.2	135.0	7.5989	X	X		X	X		X	X		X	X	X	X	X	X		X
14	71.3	71.3	71.3	38.1	7.5878	X	X		X	X		X	X		X	X	X	X	X	X	X	X
14	71.3	71.3	71.3	40.7	7.5881	X	X		X	X		X	X		X	X	X	X	X		X	X
15	71.4	71.3	71.3	27.8	7.5865	X	X		X	X		X	X		X	X	X	X	X	X	X	X
15	71.3	71.3	71.3	33.0	7.5871	X	X		X	X		X	X		X	X	X	X	X	X		X
16	71.4	71.3	71.3	22.7	7.5858	X	X		X	X		X	X		X	X	X	X	X	X	X	X
16	71.4	71.3	71.3	24.1	7.5860	X	X		X	X		X	X		X	X	X	X	X	X	X	X
17	71.4	71.3	71.3	18.8	7.5853	X	X		X	X		X	X		X	X	X	X	X	X	X	X
17	71.4	71.3	71.3	23.9	7.5858	X	X		X	X		X	X		X	X	X	X	X	X	X	X
18	71.4	71.3	71.3	19.0	7.5852	X	X		X	X		X	X		X	X	X	X	X	X	X	X

Analysis:

Interestingly, the best subsets analysis reveals that the most effective model includes all of the predictors, so no changes should be made. This best model has been determined based on three factors. The R-squared value of 71.4% is tied for the largest R-squared. The Mallows Cp is technically perfect as it is exactly one more than the number of predictors included. Additionally, the value of S for this model is the lowest of all subsets. This is surprising, as the matrix plot and correlation matrix previously demonstrated the existence of multicollinearity. However, this could also be a sign that the multicollinearity is less serious than previously interpreted, as removing some or all of the correlated variables does not benefit the model.

Hypothesis Tests and Conclusions with the Refined Model:

For convenience in naming, variables are indicated with a numerical value from one to eighteen, in the order which they are presented in the regression information

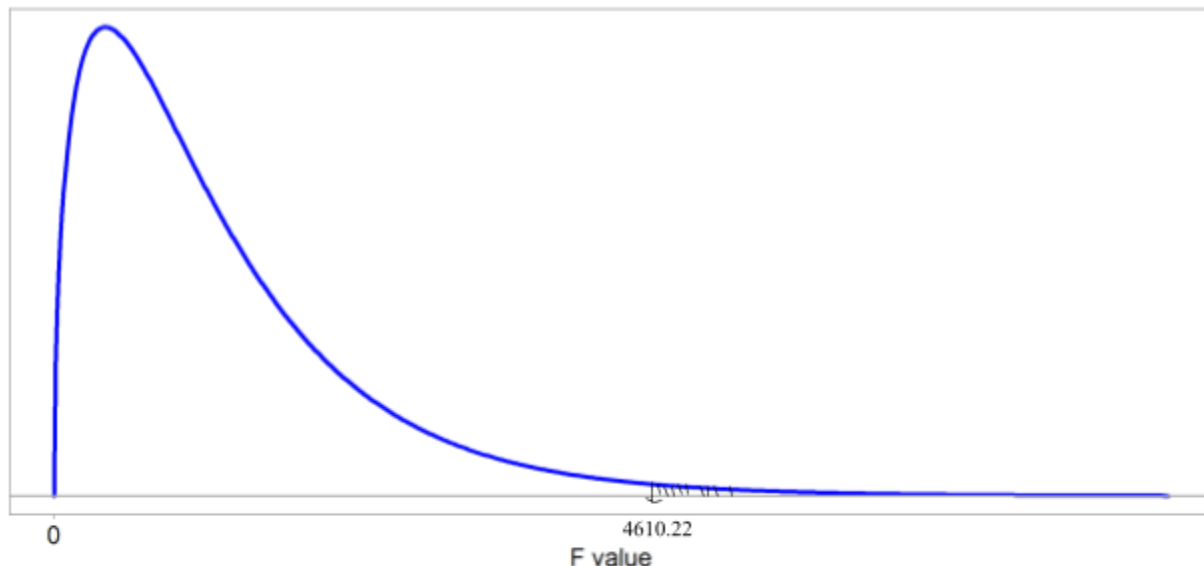
ANOVA F Test for Cumulative Effect:

$H_0: B_1=B_2=B_3=B_4=B_5=B_6=B_7=B_8=B_9=B_{10}=B_{11}=B_{12}=B_{13}=B_{14}=B_{15}=B_{16}=B_{17}=B_{18}=0$

$H_a: \text{At least one } B_j \neq 0$

$DF = (p, n - p - 1) = (18, 33,316 - 18 - 1) = (18, 33,297)$

Test-Statistic: $F = 4610.22$



P-value: $p = 0.000$

$p \leq \alpha?$ $0.000 \leq 0.05$ ✓ **Do Reject H_0**

We do have sufficient evidence to support the alternative hypothesis that at least one of field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn influence point differential.

Assessing the Effect of Individual Predictors With T-Tests and Confidence Intervals:

Before completing any statistical analysis, I predicted that field goals made, offensive rebounds, and defensive rebounds would have the strongest positive correlation with point differential and that turnovers, personal fouls, and blocks against would have the strongest negative correlation. For each of these variables, a t-test for the slope and a confidence interval for the slope are completed, which should be able to provide insight into the accuracy of my predictions.

Field Goals Made:

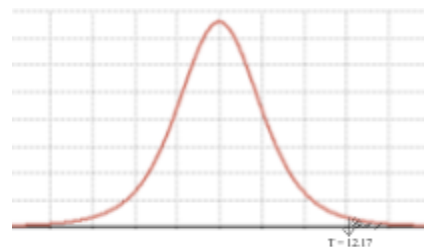
$H_0: B_1=0$ $H_a: B_1>0$

$DF = (n - p - 1) = (33316 - 18 - 1) = 33,297$

Test-Statistic: $t = 12.17$

P-value: $p = 0.000$

$p \leq \alpha?$ $0.000 \leq 0.05$ ✓ **Do Reject H_0**



We do have sufficient evidence to support the alternative hypothesis that field goals made positively influences (positive slope) point differential when field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws

made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are included in the model.

95% Confidence Interval $\rightarrow df = 33,297 \rightarrow t^* = 1.962$ (estimation with $df = 1000$)

$$b_1 = 1.2142 \quad SE_{b_1} = 0.0998$$

$$1.2142 \pm (1.962) \cdot (0.0998)$$

$$1.2142 \pm 0.1958$$

$$(1.0184, 1.4100)$$

We are 95 percent confident that for every increase by one in field goals made, we can expect an increase of between 1.0184 and 1.4100 points to the point differential when field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are fixed.

We can additionally conclude that increasing the number of field goals made always increases the point differential because all the values included in the confidence interval are positive.

Offensive Rebounds:

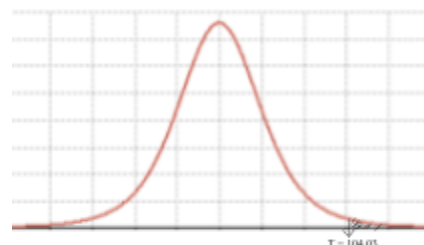
$$H_0: B_{10} = 0 \quad H_a: B_{10} > 0$$

$$DF = (n - p - 1) = (33316 - 18 - 1) = \mathbf{33,297}$$

$$\text{Test-Statistic: } t = 104.03$$

$$\text{P-value: } p = 0.000$$

$$p \leq \alpha? \quad 0.000 \leq 0.05 \quad \checkmark \quad \text{Do Reject } H_0$$



We do have sufficient evidence to support the alternative hypothesis that offensive rebounds positively influences (positive slope) point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, defensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are included in the model.

95% Confidence Interval $\rightarrow df = 33,297 \rightarrow t^* = 1.962$ (estimation with $df = 1000$)

$b_{10} = 1.5217$ $SE_{b_{10}} = 0.0146$

$1.5217 \pm (1.962) * (0.0146)$

1.5217 ± 0.0286

$(1.4931, 1.5503)$

We are 95 percent confident that for every increase by one in offensive rebounds, we can expect an increase of between 1.4931 and 1.5503 points to the point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, defensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are fixed.

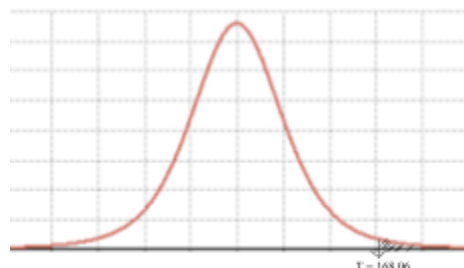
We can additionally conclude that increasing the number of offensive rebounds always increases the point differential because all the values included in the confidence interval are positive.

Defensive Rebounds:

$H_0: B_{11} = 0$ $H_a: B_{11} > 0$

$DF = (n - p - 1) = (33316 - 18 - 1) = 33,297$

Test-Statistic: $t = 168.06$



P-value: $p = 0.000$

$p \leq \alpha?$ $0.000 \leq 0.05$ ✓ **Do Reject H_0**

We do have sufficient evidence to support the alternative hypothesis that defensive rebounds positively influences (positive slope) point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are included in the model.

95% Confidence Interval $\rightarrow df = 33,297 \rightarrow t^* = 1.962$ (estimation with $df = 1000$)

$b_{11} = 1.4581$ $SE_{b_{11}} = 0.0087$

$1.4581 \pm (1.962) * (0.0087)$

1.4581 ± 0.0171

$(1.4410, 1.4752)$

We are 95 percent confident that for every increase by one in defensive rebounds, we can expect an increase of between 1.4410 and 1.4752 points to the point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, assists, turnovers, steals, blocks, blocks against, personal fouls, and personal fouls drawn are fixed.

We can additionally conclude that increasing the number of defensive rebounds always increases the point differential because all the values included in the confidence interval are positive.

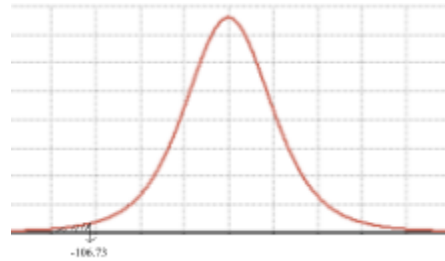
Turnovers:

$$H_0: B_{13}=0 \quad H_a: B_{13}<0$$

$$DF = (n - p - 1) = (33316 - 18 - 1) = \mathbf{33,297}$$

$$\text{Test-Statistic: } t = -106.73$$

$$\text{P-value: } p = 0.000$$



$$p \leq \alpha? \quad 0.000 \leq 0.05 \quad \checkmark \quad \mathbf{Do \text{ Reject } H_0}$$

We do have sufficient evidence to support the alternative hypothesis that turnovers negatively influences (negative slope) point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, steals, blocks, blocks against, personal fouls, and personal fouls drawn are included in the model.

$$95\% \text{ Confidence Interval} \rightarrow df = 33,297 \rightarrow t^* = 1.962 \text{ (estimation with } df = 1000)$$

$$b_{13} = -1.3649 \quad SE_{b_{13}} = 0.0128$$

$$-1.3649 \pm (1.962) \cdot (0.0128)$$

$$-1.3649 \pm 0.0251$$

$$(-1.3900, -1.3398)$$

We are 95 percent confident that for every increase by one in turnovers, we can expect a decrease of between 1.3900 and 1.3398 points to the point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, steals, blocks, blocks against, personal fouls, and personal fouls drawn are fixed.

We can additionally conclude that increasing the number of turnovers always decreases the point differential because all the values included in the confidence interval are negative.

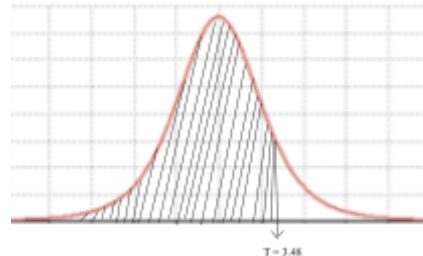
Personal Fouls:

$$H_0: B_{17}=0 \quad H_a: B_{17}<0$$

$$DF = (n - p - 1) = (33316 - 18 - 1) = \mathbf{33,297}$$

$$\text{Test-Statistic: } t = 3.48$$

$$\text{P-value: } p = 1 - P(T > 3.48) = 1 - 0.001 = 0.999$$



$$p \leq \alpha? \quad 0.999 \not\leq 0.05 \quad \mathbf{Do\ Not\ Reject\ H_0}$$

We do not have sufficient evidence to support the alternative hypothesis that personal fouls negatively influences (negative slope) point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, blocks against, and personal fouls drawn are included in the model.

$$95\% \text{ Confidence Interval} \rightarrow df = 33,297 \rightarrow t^* = 1.962 \text{ (estimation with } df = 1000)$$

$$b_{17} = 0.0364 \quad SE_{b_{17}} = 0.0104$$

$$0.0364 \pm (1.962) \cdot (0.0104)$$

$$0.0364 \pm 0.0204$$

$$(0.0160, 0.0568)$$

We are 95 percent confident that for every increase by one in personal fouls, we can expect an increase of between 0.0160 and 0.0568 points to the point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive

rebounds, defensive rebounds, assists, turnovers, steals, blocks, blocks against, and personal fouls drawn are fixed.

We can additionally conclude that increasing the number of personal fouls always increases the point differential because all the values included in the confidence interval are positive. This is the very opposite of my prediction, and the reason for this is to be disputed.

Blocks Against:

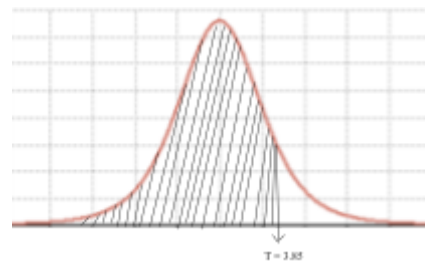
$$H_0: B_{16}=0 \quad H_a: B_{16}<0$$

$$DF = (n - p - 1) = (33316 - 18 - 1) = \mathbf{33,297}$$

$$\text{Test-Statistic: } t = 3.85$$

$$\text{P-value: } p = 1 - P(T > 3.85) = 1 - \approx 0 = \approx 1.000$$

$$p \leq \alpha? \quad \approx 1.000 \not\leq 0.05 \quad \mathbf{\text{Do Not Reject } H_0}$$



We do not have sufficient evidence to support the alternative hypothesis that blocks against negatively influences (negative slope) point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, personal fouls, and personal fouls drawn are included in the model.

$$95\% \text{ Confidence Interval} \rightarrow df = 33,297 \rightarrow t^* = 1.962 \text{ (estimation with } df = 1000)$$

$$b_{16} = 0.0708 \quad SE_{b_{16}} = 0.0184$$

$$0.0708 \pm (1.962) \cdot (0.0184)$$

$$0.0708 \pm 0.0361$$

$$(0.0347, 0.1069)$$

We are 95 percent confident that for every increase by one in blocks against, we can expect an increase of between 0.0347 and 0.1069 points to the point differential when field goals made, field goals attempted, field goal percentage, three pointers made, three pointers attempted, three point percentage, free throws made, free throws attempted, free throw percentage, offensive rebounds, defensive rebounds, assists, turnovers, steals, blocks, personal fouls, and personal fouls drawn are fixed.

We can additionally conclude that increasing the number of blocks against always increases the point differential because all the values included in the confidence interval are positive. This is also the opposite of my prediction, and the reason for this is to be disputed.

Analysis:

First of all, the T-tests and confidence intervals have certainly confirmed my prediction that field goals made, offensive rebounds, and defensive rebounds have strong positive linear relationships with point differential, meaning that having a higher number in any of these categories will increase the chance of a positive point differential and a successful game. Next, we see that I was correct about turnovers - they have a very negative correlation with point differential. Even by simply looking at the magnitude of the t-value (-106.73) we can see that the linear relationship is going to be very negative. This aligns perfectly with basketball intuition, as limiting turnovers is one of the most emphasized parts of the game. Finally, my predictions about personal fouls and blocks against were shown to be incorrect. This is very surprising, as getting a personal foul can sometimes give the opposing team easy points at the free throw line, and getting blocked can quickly lead to a counter-attack in transition - also resulting in quick and easy points. There are two possible explanations for this issue. The simple explanation would be that my predictions are

correct in actuality, but the multicollinearity of some of the predictor variables may alter the accuracy of the demonstrated individual effects of the variables. The second explanation involves the idea that after getting a block or fouling someone, which is sometimes considered embarrassing in the NBA, a player might respond to this by trying to make up for it quickly. For example, when LeBron James gets blocked, he takes it personally and runs the full length of the court to block the opponent back. Sometimes this can shift the momentum of a game and get the crowd involved, leading to a better outcome than expected.

Overall Conclusion:

After extensive analysis of the effect of various basketball statistical records on the outcome of a game, we have been able to determine many useful relationships and make powerful conclusions. The success of our multiple linear regression model is clearly excellent in some categories, but limited in others. The refined model has an R-squared value of 71.4%, which tells us that 71.4% of the variation in point differential is explained by the model. This is decent, but we would consider the model much more accurate if the R-squared value was closer to 90 - 95%. The most successful part of our model was its ability to easily pass all 5 assumptions. All of these checks were met with absolute confidence. On the more negative side, our model demonstrated large amounts of multicollinearity, which could make it difficult to establish the individual effects of predictor variables on the response. Finally, we were able to conclude that at least one of these basketball statistics does have a significant influence on the point differential of the game using the ANOVA F test, and we dove further into the individual relationships with t-tests. These t-tests had mixed results, surprisingly going against some of my initial predictions based on personal knowledge of basketball. Despite this, we have still been

able to analyze these relationships and raise the question of whether my predictions were actually based in reality. Overall, our model displays many statistical patterns and relationships, along with powerful mathematical evidence, that may continue to be inferred upon into the future.

Raw Data